Quadrature Rules and Its Application

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Chapter 1

Introduction

1.1 Background

When we need to find the area under a curve numerically, **Gauss quadrature** is one of the smartest methods available. Unlike simpler methods that use evenly spaced points, it strategically picks the 'best' points and weights to get a highly accurate answer with very few calculations. This efficiency makes it a cornerstone of advanced numerical analysis.

1.2 Concept and Derivation

Gauss quadrature's main advantage comes from treating both the sample points (**nodes**) and their **weights** as variables to be optimized. Simpler methods, like the trapezoidal rule, fix the nodes at the interval endpoints. Gaussian quadrature, however, finds the optimal placement of these nodes to achieve the highest possible accuracy for a given number of points.

The general form of any quadrature rule approximates an integral as a weighted sum:

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} w_i f(x_i)$$

Here, x_i are the nodes and w_i are the corresponding weights. With n points, we have $2n$ parameters to choose (n nodes plus n weights). Gauss quadrature leverages this to exactly integrate polynomials of degree up to $2n-1$.

1.3 Connection to Orthogonal Polynomials

The mathematical elegance of Gaussian quadrature comes from its deep connection to **orthogonal polynomials**. The optimal nodes, x_i, are precisely the roots of the n-th degree Legendre polynomial over the integration interval $[-1, 1]$. This relationship guarantees that the quadrature rule achieves its maximum theoretical precision.

For Gauss-Legendre quadrature on $[-1, 1]$, the nodes are the roots of Legendre polynomials, and the weights are calculated using the formula:

$$w_i = \frac{2}{(1 - x_i^2)[P_n'(x_i)]^2}$$

where $P_n'(x_i)$ is the derivative of the Legendre polynomial evaluated at the node x_i.

Chapter 2

Gauss–Legendre Quadrature Derivation

2.1 Exactness of Gaussian Quadrature

For $N = 2n - 1$, there exist Gaussian points x_i and weights w_i such that

$$\int_{-1}^{1} x^k \, dx = \sum_{i=1}^{n} w_i \, (x_i)^k$$

where $k = 0, 1, 2, \ldots, N$.

Proof

Let the Gaussian points x_i be the roots of the Legendre polynomial of degree n, denoted $P_n(x)$. Let $P(x)$ be an arbitrary polynomial with degree $\leq 2n - 1$.

By the Division Algorithm for Polynomials, we can write $P(x)$ as:

$$P(x) = P_n(x)Q(x) + R(x)$$

where $Q(x)$ and $R(x)$ are polynomials with degrees $\leq n - 1$.

The definition of $P_n(x)$ based on its roots is:

$$P_n(x) = \prod_{i=1}^{n} (x - x_i)$$

Since the Gaussian points x_i are the roots, we have $P_n(x_i) = 0$. Evaluating $P(x)$ at these points yields:

$$P(x_i) = P_n(x_i)Q(x_i) + R(x_i) \implies P(x_i) = R(x_i)$$

This establishes the key relationship between the polynomial and the remainder.

Now, we integrate the polynomial $P(x)$ over the interval $[-1, 1]$:

$$\begin{aligned}\int_{-1}^{1} P(x)\,dx &= \int_{-1}^{1} [P_n(x)Q(x) + R(x)]\,dx \\ &= \int_{-1}^{1} P_n(x)Q(x)\,dx + \int_{-1}^{1} R(x)\,dx\end{aligned}$$

Since $P_n(x)$ is a Legendre polynomial (orthogonal with respect to the weight function $w(x) = 1$), and $\deg(Q) \leq n - 1$, the **Orthogonality property** ensures that the first integral is zero:

$$\int_{-1}^{1} P_n(x)Q(x)\,dx = 0$$

Therefore, the integral simplifies to:

$$\int_{-1}^{1} P(x)\,dx = \int_{-1}^{1} R(x)\,dx$$

The polynomial $R(x)$, with $\deg(R) \leq n - 1$, can be exactly integrated using the n-point quadrature rule because the weights w_i are chosen to exactly integrate all polynomials of degree $\leq n - 1$. Specifically, $R(x)$ can be expressed using the Lagrange basis polynomials $L_i(x)$:

$$\begin{aligned}\int_{-1}^{1} R(x)\,dx &= \int_{-1}^{1} \sum_{i=1}^{n} R(x_i)L_i(x)\,dx \\ &= \sum_{i=1}^{n} R(x_i) \int_{-1}^{1} L_i(x)\,dx \\ &= \sum_{i=1}^{n} R(x_i)w_i\end{aligned}$$

Finally, substituting $R(x_i) = P(x_i)$ from above, we get:

$$\int_{-1}^{1} P(x)\,dx = \sum_{i=1}^{n} P(x_i)w_i$$

This proves that the n-point Gaussian Quadrature rule is exact for an arbitrary polynomial $P(x)$ with degree $\leq 2n - 1$.

2.2 2-Point Rule

For the 2-point rule over the canonical interval $[-1, 1]$, the approximation is:

$$\int_{-1}^{1} f(x)\,dx \approx w_1 f(x_1) + w_2 f(x_2) \tag{2.1}$$

To find the four unknown parameters (w_1, w_2, x_1, x_2), we enforce that the rule is exact for polynomials of degree up to $2n - 1 = 3$. This gives us a system of four equations using the basis functions $1, x, x^2, x^3$.

2.2.1 System of Governing Equations

$$f(x) = 1: \quad \int_{-1}^{1} 1\,dx = 2 \quad \Longrightarrow \quad w_1 + w_2 = 2 \tag{2.2}$$

$$f(x) = x: \quad \int_{-1}^{1} x\,dx = 0 \quad \Longrightarrow \quad w_1 x_1 + w_2 x_2 = 0 \tag{2.3}$$

$$f(x) = x^2: \quad \int_{-1}^{1} x^2\,dx = \frac{2}{3} \quad \Longrightarrow \quad w_1 x_1^2 + w_2 x_2^2 = \frac{2}{3} \tag{2.4}$$

$$f(x) = x^3: \quad \int_{-1}^{1} x^3\,dx = 0 \quad \Longrightarrow \quad w_1 x_1^3 + w_2 x_2^3 = 0 \tag{2.5}$$

Due to the symmetry of the interval $[-1, 1]$, we can assume $x_1 = -x_2$ and $w_1 = w_2$. From (2.2), substituting $w_1 = w_2$:

$$\begin{aligned} 2w_1 = 2 &\Longrightarrow w_1 = 1 \\ \therefore w_2 &= 1 \end{aligned}$$

From (2.4), substituting the weights and $x_1 = -x_2$:

$$\begin{aligned} x_1^2 + (-x_1)^2 &= \frac{2}{3} \\ \Longrightarrow 2x_1^2 &= \frac{2}{3} \\ \Longrightarrow x_1 &= \pm\frac{1}{\sqrt{3}} \end{aligned}$$

We choose $x_1 = -1/\sqrt{3}$ and $x_2 = 1/\sqrt{3}$.

2.2.2 Final 2-Point Formula

Substituting the nodes and weights into (2.1):

$$\int_{-1}^{1} f(x)\,dx \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

2.3 3-Point Rule

The integral approximation for the 3-point rule is given by:

$$\int_{-1}^{1} f(x)\,dx \approx w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)$$

Here we have 6 unknowns (3 nodes, 3 weights), allowing for exactness up to degree $2n - 1 = 5$.

2.3.1 System of Equations

The formula must be exact for $f(x) = 1, x, x^2, x^3, x^4, x^5$:

$$f(x) = 1: \quad \int_{-1}^{1} 1\,dx = 2 = w_1 + w_2 + w_3 \tag{2.6}$$

$$f(x) = x: \quad \int_{-1}^{1} x\,dx = 0 = w_1 x_1 + w_2 x_2 + w_3 x_3 \tag{2.7}$$

$$f(x) = x^2: \quad \int_{-1}^{1} x^2\,dx = \frac{2}{3} = w_1 x_1^2 + w_2 x_2^2 + w_3 x_3^2 \tag{2.8}$$

$$f(x) = x^3: \quad \int_{-1}^{1} x^3\,dx = 0 = w_1 x_1^3 + w_2 x_2^3 + w_3 x_3^3 \tag{2.9}$$

$$f(x) = x^4: \quad \int_{-1}^{1} x^4\,dx = \frac{2}{5} = w_1 x_1^4 + w_2 x_2^4 + w_3 x_3^4 \tag{2.10}$$

$$f(x) = x^5: \quad \int_{-1}^{1} x^5\,dx = 0 = w_1 x_1^5 + w_2 x_2^5 + w_3 x_3^5 \tag{2.11}$$

2.3.2 Solving with Symmetry

By symmetry of the interval, we assume: $x_2 = 0$, $x_3 = -x_1$, and $w_3 = w_1$. This simplifies the system significantly.

Finding the Nodes (x_i)

The even-powered equations simplify:

$$\text{From (2.6):} \quad 2w_1 + w_2 = 2 \tag{2.12}$$

$$\text{From (2.8):} \quad 2w_1 x_1^2 = \frac{2}{3} \tag{2.13}$$

$$\text{From (2.10):} \quad 2w_1 x_1^4 = \frac{2}{5} \tag{2.14}$$

Dividing (2.14) by (2.13) to eliminate w_1:

$$\frac{2w_1 x_1^4}{2w_1 x_1^2} = \frac{2/5}{2/3}$$

$$x_1^2 = \frac{3}{5} \implies x_1 = \pm\sqrt{\frac{3}{5}}$$

Therefore, the nodes are: $x_1 = -\sqrt{\frac{3}{5}}$, $x_2 = 0$, and $x_3 = \sqrt{\frac{3}{5}}$.

Finding the Weights (w_i)

Substitute $x_1^2 = 3/5$ into (2.13):

$$2w_1\left(\frac{3}{5}\right) = \frac{2}{3} \implies w_1 = \frac{5}{9}$$

Since $w_3 = w_1$, we have $w_3 = 5/9$.

Finally, use the simplest equation, $2w_1 + w_2 = 2$, to find w_2:

$$\begin{aligned} 2\left(\frac{5}{9}\right) + w_2 &= 2 \\ \frac{10}{9} + w_2 = 2 &\implies w_2 = 2 - \frac{10}{9} = \frac{8}{9} \end{aligned}$$

2.3.3 Final 3-Point Formula

Substituting the nodes and weights gives the final formula:

$$\int_{-1}^{1} f(x)\,dx \approx \frac{5}{9}f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9}f(0) + \frac{5}{9}f\left(\sqrt{\frac{3}{5}}\right)$$

2.4 Error and Accuracy

If $f \in O^{2n}([-1,1])$, then there exists $\xi \in (-1,1)$ such that

$$E_n(f) = \int_{-1}^{1} f(x)\,dx - \sum_{i=1}^{n} w_i f(x_i) = C_n f^{(2n)}(\xi),$$

where the constant C_n is

$$C_n = \frac{2^{2n+1}(n!)^4}{(2n+1)((2n)!)^3}.$$

So the error is proportional to the $2n$-th derivative evaluated at some point in the interval.

Numeric values of C_n (for small n) are:

- $n = 1$: $C_1 = \frac{1}{3} \approx 0.333333$
- $n = 2$: $C_2 \approx 0.0074074074$
- $n = 3$: $C_3 \approx 6.3492 \times 10^{-5}$
- $n = 4$: $C_4 \approx 2.8795 \times 10^{-7}$

2.5 Code for Gaussian Quadrature

```python
import numpy as np
from numpy.linalg import eigh

def gauss_legendre_golub_welsch(n):

    diag_a = np.zeros(n)
    k = np.arange(1, n)
    beta = k**2 / (4 * k**2 - 1)
    diag_b = np.sqrt(beta)
    J = np.diag(diag_a) + np.diag(diag_b, k=1) + np.diag(diag_b, k=-1)
    eigenvalues, eigenvectors = eigh(J)
    nodes = eigenvalues
    mu_0 = 2.0
    weights = mu_0 * (eigenvectors[0, :])**2
    return nodes, weights

n = 3
x, w = gauss_legendre_golub_welsch(n)

print(f"Number of points (n): {n}")
print(f"Nodes (x_i): {x}")
print(f"Weights (w_i): {w}")
```

Listing 2.1: Golub-Welsch Algorithm for Gauss-Legendre Quadrature

n	Nodes (x_i)		Weights (W_i)	
	Exact Form	**Decimal (10 Places)**	**Exact Form**	**Decimal (10 Places)**
1	0	0.0000000000	2	2.0000000000
2	$\pm\frac{1}{\sqrt{3}}$	$\pm$0.5773502692	1	1.0000000000
3	0	0.0000000000	$\frac{8}{9}$	0.8888888889
	$\pm\sqrt{\frac{3}{5}}$	$\pm$0.7745966692	$\frac{5}{9}$	0.5555555556
4	$\pm\sqrt{\frac{3}{7}-\frac{2}{7}\sqrt{\frac{6}{5}}}$	$\pm$0.3399810436	$\frac{18+\sqrt{30}}{36}$	0.6521451549
	$\pm\sqrt{\frac{3}{7}+\frac{2}{7}\sqrt{\frac{6}{5}}}$	$\pm$0.8611363116	$\frac{18-\sqrt{30}}{36}$	0.3478548451
5	0	0.0000000000	$\frac{128}{225}$	0.5688888889
	$\pm\frac{1}{3}\sqrt{5-2\sqrt{\frac{10}{7}}}$	$\pm$0.5384693101	$\frac{322+13\sqrt{70}}{900}$	0.4786286705
	$\pm\frac{1}{3}\sqrt{5+2\sqrt{\frac{10}{7}}}$	$\pm$0.9061798459	$\frac{322-13\sqrt{70}}{900}$	0.2369268851

Table 2.1: Nodes (x_i) and Weights (W_i) for n-Point Gauss–Legendre Rules (Exact vs. Decimal)

Chapter 3

Radau Quadrature

The Gauss-Radau quadrature rule is a variation of Gaussian quadrature where one of the endpoints of the integration interval $[-1, 1]$ is required to be a node. Assuming the left endpoint, $\mathbf{x_1 = -1}$, is fixed:

$$\int_{-1}^{1} f(x)\,dx = w_1 f(-1) + \sum_{i=2}^{n} w_i f(x_i) \tag{3.1}$$

For n total nodes, there are $2n - 1$ unknowns ($n - 1$ variable nodes, n weights). This structure allows the rule to be exact for polynomials of degree up to $\mathbf{2n - 2}$.

3.1 3-Point Rule ($n = 3$)

For $n = 3$, the rule is exact for polynomials up to degree $2n - 2 = 4$. We have five unknown parameters: w_1, w_2, w_3, x_2, x_3. The approximation is:

$$\int_{-1}^{1} f(x)\,dx = w_1 f(-1) + w_2 f(x_2) + w_3 f(x_3) \tag{3.2}$$

3.1.1 System of Governing Equations

We enforce exactness for the basis functions $f(x) = 1, x, x^2, x^3, x^4$:

$$f(x) = 1: \quad \int_{-1}^{1} 1\,dx = 2 \quad \Longrightarrow \quad w_1 + w_2 + w_3 = 2 \tag{3.3}$$

$$f(x) = x: \quad \int_{-1}^{1} x\,dx = 0 \quad \Longrightarrow \quad -w_1 + w_2 x_2 + w_3 x_3 = 0 \tag{3.4}$$

$$f(x) = x^2: \quad \int_{-1}^{1} x^2\,dx = \frac{2}{3} \quad \Longrightarrow \quad w_1 + w_2 x_2^2 + w_3 x_3^2 = \frac{2}{3} \tag{3.5}$$

$$f(x) = x^3: \quad \int_{-1}^{1} x^3\,dx = 0 \quad \Longrightarrow \quad -w_1 + w_2 x_2^3 + w_3 x_3^3 = 0 \tag{3.6}$$

$$f(x) = x^4: \quad \int_{-1}^{1} x^4\,dx = \frac{2}{5} \quad \Longrightarrow \quad w_1 + w_2 x_2^4 + w_3 x_3^4 = \frac{2}{5} \tag{3.7}$$

3.1.2 Solution

The nodes x_2, x_3 are the roots of the polynomial $P_3(x) - P_2(x)$, where $P_k(x)$ is the Legendre polynomial of degree k. The solutions are found to be:

$$x_1 = -1, \quad x_2 = \frac{1-\sqrt{6}}{5}, \quad x_3 = \frac{1+\sqrt{6}}{5}$$

The corresponding weights are:

$$w_1 = \frac{2}{9}, \quad w_2 = \frac{16+\sqrt{6}}{18}, \quad w_3 = \frac{16-\sqrt{6}}{18}$$

3.1.3 Final 3-Point Formula

Substituting the nodes and weights into (3.2) yields the final formula:

$$\int_{-1}^{1} f(x)\,dx \approx \frac{2}{9}f(-1) + \frac{16+\sqrt{6}}{18}f\left(\frac{1-\sqrt{6}}{5}\right) + \frac{16-\sqrt{6}}{18}f\left(\frac{1+\sqrt{6}}{5}\right)$$

3.2 Summary of Nodes and Weights

The table below summarizes the Radau-Left nodes and weights, where $x_1 = -1$.

Table 3.1: Nodes (x_i) and Weights (W_i) for n-Point Radau Quadrature ($x_1 = -1$ fixed)

n	**Fixed Node**	**Variable Nodes (x_i)**	**Weights (w_i)**
2	$x_1 = -1$	$x_2 = \frac{1}{3} \approx 0.3333$	$w_1 = \frac{1}{2} = 0.5$ $w_2 = \frac{3}{2} = 1.5$
3	$x_1 = -1$	$x_2 = \frac{1-\sqrt{6}}{5} \approx -0.2899$ $x_3 = \frac{1+\sqrt{6}}{5} \approx 0.6899$	$w_1 = \frac{2}{9} \approx 0.2222$ $w_2 = \frac{16+\sqrt{6}}{18} \approx 1.0250$ $w_3 = \frac{16-\sqrt{6}}{18} \approx 0.7528$

3.3 Error and Accuracy

The error (remainder term) for the n-point Radau rule (with $x = -1$ fixed) is a function of the $(2n-1)$-th derivative of $f(x)$.

For $f \in C^{2n-1}[-1, 1]$, the error $R_n[f]$ is:

$$R_n[f] = C_n f^{(2n-1)}(\xi), \quad \xi \in (-1, 1)$$

where C_n is a constant related to the integral of the squared polynomial product over the weighted interval.

3.3.1 Actionable Insight

The crucial detail is that the error is governed by the $f^{(2n-1)}(\xi)$ term, which is why the degree of exactness is **$2n-2$**. If $f(x)$ is a polynomial of degree $2n-2$ or less, its $(2n-1)$-th derivative is zero, ensuring the quadrature is exact.

3.4 Code for Radau Quadrature

The implementation of Radau quadrature relies on the **roots_jacobi** function from `scipy.special`, as Radau nodes are roots of Jacobi polynomials with specific parameters ($\alpha = 0, \beta = 1$ for the $x = -1$ fixed case).

```python
import numpy as np
from scipy.special import roots_jacobi

def calculate_radau_quadrature(N: int, fixed_endpoint: float = -1.0):
    if N < 2:
        raise ValueError("N must be at least 2 for Radau quadrature.")
    if fixed_endpoint not in [-1.0, 1.0]:
        raise ValueError("fixed_endpoint must be -1.0 or 1.0.")

    n_variable = N - 1

    # Determine Jacobi parameters (alpha, beta) based on fixed endpoint
    if fixed_endpoint == -1.0:
        # Radau-Left: Fixed node at -1. Corresponds to Jacobi (0, 1)
        alpha = 0
        beta = 1
    else:
        # Radau-Right: Fixed node at 1. Corresponds to Jacobi (1, 0)
        alpha = 1
        beta = 0

    # Get nodes and weights for the underlying n-1 point Jacobi rule
    t, w_jacobi = roots_jacobi(n_variable, alpha, beta)

    # Transform Jacobi weights to Radau weights for the n-1 variable nodes
    if fixed_endpoint == -1.0:
        # Weights for variable nodes (x_2 to x_n)
```

```
        w_radau = 2.0 * w_jacobi / (1.0 + t)
    else:
        # Weights for variable nodes (x_1 to x_{n-1})
        w_radau = 2.0 * w_jacobi / (1.0 - t)

    # Calculate the weight for the fixed endpoint
    w_fixed = 2.0 - np.sum(w_radau)

    # Combine nodes and weights
    # Insert fixed node and weight at the correct position (0 for -1, end for 1)
    nodes = np.insert(t, 0 if fixed_endpoint == -1.0 else n_variable, fixed_endpoint)
    weights = np.insert(w_radau, 0 if fixed_endpoint == -1.0 else n_variable, w_fixed)

    # Sort nodes and weights for standard presentation
    sort_indices = np.argsort(nodes)
    return nodes[sort_indices], weights[sort_indices]

N_points = 3
nodes_3, weights_3 = calculate_radau_quadrature(N_points, fixed_endpoint=-1.0)
print(f"--- N={N_points} Radau-Left Rule (Fixed x=-1) ---")
print(f"Nodes (x_i):   {nodes_3}")
print(f"Weights (w_i): {weights_3}")
```

Listing 3.1: Code for Radau Quadrature using SciPy

Chapter 4

Lobatto Quadrature Derivation

The Gauss-Lobatto quadrature rule is a form of closed Gaussian quadrature where **both endpoints**, $\mathbf{x_1 = -1}$ and $\mathbf{x_n = 1}$, are included as fixed nodes. The general approximation form is:

$$\int_{-1}^{1} f(x)\,dx = w_1 f(-1) + w_n f(1) + \sum_{i=2}^{n-1} w_i f(x_i) \tag{4.1}$$

With n total nodes, there are $2n-2$ unknown parameters ($n-2$ variable nodes, n weights). This structure ensures the method is exact for polynomials of degree up to $\mathbf{2n-3}$.

4.1 3-Point Rule ($n = 3$)

For $n = 3$, the rule is exact for polynomials up to degree $2n - 3 = 3$. We have four unknown parameters: w_1, w_2, w_3, x_2. The approximation is:

$$\int_{-1}^{1} f(x)\,dx = w_1 f(-1) + w_3 f(1) + w_2 f(x_2) \tag{4.2}$$

4.1.1 System of Governing Equations

We enforce exactness for the basis functions $f(x) = 1, x, x^2, x^3$:

$$f(x) = 1: \quad \int_{-1}^{1} 1\,\mathrm{d}x = 2 \quad \Longrightarrow \quad w_1 + w_2 + w_3 = 2 \tag{4.3}$$

$$f(x) = x: \quad \int_{-1}^{1} x\,\mathrm{d}x = 0 \quad \Longrightarrow \quad -w_1 + w_2 x_2 + w_3 = 0 \tag{4.4}$$

$$f(x) = x^2: \quad \int_{-1}^{1} x^2\,\mathrm{d}x = \frac{2}{3} \quad \Longrightarrow \quad w_1 + w_2 x_2^2 + w_3 = \frac{2}{3} \tag{4.5}$$

$$f(x) = x^3: \quad \int_{-1}^{1} x^3\,\mathrm{d}x = 0 \quad \Longrightarrow \quad -w_1 + w_2 x_2^3 + w_3 = 0 \tag{4.6}$$

4.1.2 Solution

Due to the symmetry of the interval and moments:

1. From (4.4) and (4.6), we deduce that $w_2 x_2 = w_2 x_2^3$. Since $w_2 \neq 0$ (from (4.3)), we must have $x_2 = x_2^3$, which implies $\mathbf{x_2 = 0}$ or $x_2 = \pm 1$ (but the latter is fixed).
2. Substituting $x_2 = 0$ into (4.5): $w_1 + 0 + w_3 = \frac{2}{3}$.
3. Substituting $w_1 + w_3 = \frac{2}{3}$ into (4.3): $\frac{2}{3} + w_2 = 2 \implies \mathbf{w_2 = \frac{4}{3}}$.
4. Substituting w_2 and x_2 into (4.4) gives $-w_1 + w_3 = 0$, so $\mathbf{w_1 = w_3}$.
5. Since $w_1 + w_3 = \frac{2}{3}$, we have $2w_1 = \frac{2}{3} \implies \mathbf{w_1 = w_3 = \frac{1}{3}}$.

4.1.3 Final 3-Point Formula (Simpson's Rule)

Substituting the nodes and weights into (4.2) yields:

$$\int_{-1}^{1} f(x)\,dx \approx \frac{1}{3}f(-1) + \frac{4}{3}f(0) + \frac{1}{3}f(1) = \frac{1}{3}\left[f(-1) + 4f(0) + f(1)\right]$$

4.2 Summary of Nodes and Weights

Table 4.1: Nodes (x_i) and Weights (W_i) for n-Point Lobatto Quadrature

n	**Fixed Nodes**	**Variable Nodes (x_i)**	**Weights (w_i)**
3	± 1	0	$w_1, w_3 = 1/3 \approx 0.3333$ $w_2 = 4/3 \approx 1.3333$
4	± 1	$x_2, x_3 = \pm\sqrt{\frac{1}{5}} \approx \pm 0.44721$	$w_1, w_4 = 1/6 \approx 0.1667$ $w_2, w_3 = 5/6 \approx 0.8333$
5	± 1	$x_2, x_4 = \pm\sqrt{\frac{3}{7}} \approx \pm 0.65465$ $x_3 = 0$	$w_1, w_5 = 1/10 = 0.1000$ $w_2, w_4 = 49/90 \approx 0.5444$ $w_3 = 32/45 \approx 0.7111$

4.3 Error and Accuracy

The error (or remainder term), $E_n(f)$, for the n-point Lobatto rule is structurally tied to the $2n - 2$ derivative of the integrand $f(x)$.

4.3.1 Mean-Value Form

For a smooth function $f \in C^{2n-2}([-1,1])$, the error is:

$$E_n(f) = C_n^{(L)} f^{(2n-2)}(\xi), \quad \xi \in (-1,1)$$

where $C_n^{(L)}$ is a constant determined by the specific Lobatto rule, and ξ is some point within the integration interval.

4.3.2 Actionable Insight

The degree of exactness is $\mathbf{2n-3}$. Since the error is proportional to the $(\mathbf{2n-2})$-th derivative, if $f(x)$ is a polynomial of degree $2n-3$ or less, $f^{(2n-2)}(x)$ is zero, and the quadrature yields the exact result.

4.4 Code for Lobatto Quadrature

The Lobatto nodes are the roots of the Legendre polynomial of degree $n-1$, $P_{n-1}(x)$, plus the endpoints ± 1. The variable nodes (x_2 to x_{n-1}) are the roots of the derivative of the Legendre polynomial, $P'_{n-1}(x)$, which are related to the Jacobi polynomials $P_{n-2}^{(1,1)}(x)$. The `scipy.special.roots_jacobi` function handles this efficiently.

```python
import numpy as np
from scipy.special import roots_jacobi

def calculate_lobatto_quadrature(N: int):
    # N is the total number of nodes (including -1 and 1).
    if N < 2:
        raise ValueError("N must be at least 2 for Lobatto quadrature.")

    if N == 2:
        # Trivial case: Trapezoidal rule
        nodes = np.array([-1.0, 1.0])
        weights = np.array([1.0, 1.0])
        return nodes, weights

    # Lobatto variable nodes are roots of Jacobi P_{N-2}^{(1,1)}(x)
    n_variable = N - 2
    alpha = 1
    beta = 1

    # Get the internal nodes (t) and their weights (w_jacobi)
    t, w_jacobi = roots_jacobi(n_variable, alpha, beta)

```

```python
    # Calculate the weight for the fixed endpoints (w_endpoints)
    # The general formula for the endpoint weights is w_1 = w_n = 2 / (N*(N-1))
    w_endpoints = 2.0 / (N * (N - 1))

    # Calculate the weights for the internal nodes
    # w_i = (2 / (N*(N-1))) * (1 - t_i^2) * P_{N-2}^{(1, 1)}(t_i)
    # The weights returned by roots_jacobi for (1, 1) are w_jacobi,
    # which need scaling to get the actual w_internal
    w_internal = w_jacobi * (2.0 / ((N * (N-1)))) # Apply the correct scaling

    # Combine nodes and weights: [-1.0, t_2, ..., t_{n-1}, 1.0]
    nodes = np.concatenate(([-1.0], t, [1.0]))
    weights = np.concatenate(([w_endpoints], w_internal, [w_endpoints]))

    return nodes, weights

N_points = 3 # This will calculate the N=3 rule (3 nodes: -1, 0, 1)
nodes, weights = calculate_lobatto_quadrature(N_points)

print(f"--- N={N_points} Lobatto Rule (Fixed x=-1, x=1) ---")
print(f"Nodes (x_i):   {nodes}")
print(f"Weights (w_i): {weights}")
```

Listing 4.1: Code for Lobatto Quadrature using SciPy

Chapter 5

Gauss–Hermite Quadrature Derivation

The Gauss-Hermite quadrature rule is a powerful numerical technique for approximating integrals over the infinite interval $(-\infty, \infty)$ where the integrand includes a standard Gaussian weighting function, $w(x) = e^{-x^2}$.

$$\int_{-\infty}^{\infty} e^{-x^2} f(x)\, dx \approx \sum_{i=1}^{n} w_i f(x_i)$$

The nodes (x_i) are the roots of the Hermite polynomial $H_n(x)$, and the weights (w_i) are determined such that the rule is exact for polynomials $f(x)$ up to degree $2n-1$.

5.1 2-Point Rule ($n = 2$)

For the $n = 2$ rule, we determine four unknowns (w_1, w_2, x_1, x_2) by enforcing exactness for polynomials of degree up to $2n - 1 = 3$. We use the basis functions $1, x, x^2, x^3$.

5.1.1 System of Governing Equations

We use the known exact values for the moments $\int_{-\infty}^{\infty} x^k e^{-x^2} dx$:

$$f(x) = 1: \quad \int_{-\infty}^{\infty} 1 e^{-x^2}\, dx = \sqrt{\pi} \quad \Longrightarrow \quad w_1 + w_2 = \sqrt{\pi} \tag{5.1}$$

$$f(x) = x: \quad \int_{-\infty}^{\infty} x e^{-x^2}\, dx = 0 \quad \Longrightarrow \quad w_1 x_1 + w_2 x_2 = 0 \tag{5.2}$$

$$f(x) = x^2: \quad \int_{-\infty}^{\infty} x^2 e^{-x^2}\, dx = \frac{\sqrt{\pi}}{2} \quad \Longrightarrow \quad w_1 x_1^2 + w_2 x_2^2 = \frac{\sqrt{\pi}}{2} \tag{5.3}$$

Due to the symmetry of the Gaussian weight function, we can assume $x_1 = -x_2$ and $w_1 = w_2$.

From (5.1), substituting $w_1 = w_2$:

$$2w_1 = \sqrt{\pi} \implies w_1 = \frac{\sqrt{\pi}}{2}$$
$$\therefore w_2 = \frac{\sqrt{\pi}}{2}$$

From (5.3), substituting the weights and $x_1 = -x_2$:

$$\begin{aligned} w_1 x_1^2 + w_1 x_1^2 &= \frac{\sqrt{\pi}}{2} \\ 2w_1 x_1^2 &= \frac{\sqrt{\pi}}{2} \\ 2\left(\frac{\sqrt{\pi}}{2}\right) x_1^2 &= \frac{\sqrt{\pi}}{2} \\ \implies x_1^2 = \frac{1}{2} &\implies x_1 = \pm\frac{1}{\sqrt{2}} \end{aligned}$$

We choose $x_1 = -1/\sqrt{2}$ and $x_2 = 1/\sqrt{2}$.

5.1.2 Final 2-Point Formula

Substituting the nodes and weights into the general approximation formula:

$$\int_{-\infty}^{\infty} e^{-x^2} f(x)\, dx \approx \frac{\sqrt{\pi}}{2} f\left(-\frac{1}{\sqrt{2}}\right) + \frac{\sqrt{\pi}}{2} f\left(\frac{1}{\sqrt{2}}\right)$$

5.2 3-Point Rule ($n = 3$)

The 3-point rule involves 6 unknowns, allowing for exactness up to degree $2n - 1 = 5$. The integral approximation is:

$$\int_{-\infty}^{\infty} e^{-x^2} f(x)\, dx \approx w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)$$

5.2.1 System of Governing Equations

The formula must be exact for $f(x) = 1, x, x^2, x^3, x^4, x^5$:

$$f(x) = 1: \quad w_1 + w_2 + w_3 = \sqrt{\pi} \tag{5.4}$$
$$f(x) = x: \quad w_1 x_1 + w_2 x_2 + w_3 x_3 = 0 \tag{5.5}$$
$$f(x) = x^2: \quad w_1 x_1^2 + w_2 x_2^2 + w_3 x_3^2 = \frac{\sqrt{\pi}}{2} \tag{5.6}$$
$$f(x) = x^4: \quad w_1 x_1^4 + w_2 x_2^4 + w_3 x_3^4 = \frac{3\sqrt{\pi}}{4} \tag{5.7}$$

(Equations for $f(x) = x^3$ and $f(x) = x^5$ also equal zero and are redundant under symmetry.)

5.2.2 Solving with Symmetry

We assume symmetry: $\mathbf{x_2} = \mathbf{0}$, $\mathbf{x_3} = -\mathbf{x_1}$, and $\mathbf{w_3} = \mathbf{w_1}$.

Finding the Nodes (x_i)

The even-powered equations simplify using $x_2 = 0$ and $w_1 = w_3$:

$$\text{From (5.6):} \quad 2w_1x_1^2 = \frac{\sqrt{\pi}}{2} \tag{5.8}$$

$$\text{From (5.7):} \quad 2w_1x_1^4 = \frac{3\sqrt{\pi}}{4} \tag{5.9}$$

Dividing (5.9) by (5.8) to find x_1^2:

$$\begin{aligned} \frac{2w_1x_1^4}{2w_1x_1^2} &= \frac{3\sqrt{\pi}/4}{\sqrt{\pi}/2} \\ x_1^2 &= \frac{3}{4} \cdot 2 = \frac{3}{2} \\ \implies x_1 &= \pm\sqrt{\frac{3}{2}} \end{aligned}$$

The nodes are: $\mathbf{x_1} = -\sqrt{\mathbf{3/2}}$, $\mathbf{x_2} = \mathbf{0}$, and $\mathbf{x_3} = \sqrt{\mathbf{3/2}}$.

Finding the Weights (w_i)

Substitute $x_1^2 = 3/2$ into (5.8):

$$\begin{aligned} 2w_1\left(\frac{3}{2}\right) &= \frac{\sqrt{\pi}}{2} \\ 3w_1 = \frac{\sqrt{\pi}}{2} &\implies w_1 = \frac{\sqrt{\pi}}{6} \end{aligned}$$

Since $w_3 = w_1$, we have $\mathbf{w_3} = \sqrt{\pi}/\mathbf{6}$.

Finally, use (5.4) to find w_2:

$$\begin{aligned} 2w_1 + w_2 &= \sqrt{\pi} \\ 2\left(\frac{\sqrt{\pi}}{6}\right) + w_2 &= \sqrt{\pi} \\ w_2 &= \sqrt{\pi} - \frac{\sqrt{\pi}}{3} = \frac{2\sqrt{\pi}}{3} \end{aligned}$$

The weights are: $\mathbf{w_1} = \sqrt{\pi}/\mathbf{6}$, $\mathbf{w_2} = \mathbf{2}\sqrt{\pi}/\mathbf{3}$, and $\mathbf{w_3} = \sqrt{\pi}/\mathbf{6}$.

5.2.3 Final 3-Point Formula

$$\int_{-\infty}^{\infty} e^{-x^2} f(x)\,dx \approx \frac{\sqrt{\pi}}{6} f\left(-\sqrt{\frac{3}{2}}\right) + \frac{2\sqrt{\pi}}{3} f(0) + \frac{\sqrt{\pi}}{6} f\left(\sqrt{\frac{3}{2}}\right)$$

5.3 Summary of Nodes and Weights

n	Nodes (x_i)		Weights (W_i)	
	Exact Form	**Decimal (5 Places)**	**Exact Form**	**Decimal (5 Places)**
1	0	0.00000	$\sqrt{\pi}$	1.77245
2	$\pm\frac{1}{\sqrt{2}}$	± 0.70711	$\frac{\sqrt{\pi}}{2}$	0.88623
3	0	0.00000	$\frac{2\sqrt{\pi}}{3}$	1.18164
	$\pm\sqrt{\frac{3}{2}}$	± 1.22474	$\frac{\sqrt{\pi}}{6}$	0.29541

Table 5.1: Nodes (x_i) and Weights (W_i) for n-Point Gauss–Hermite Rules

5.4 Error and Accuracy

If $f \in O^{2n}(\mathbb{R})$, the error (remainder term) for the n-point Gauss-Hermite rule is proportional to the $2n$-th derivative evaluated at some point ξ:

$$R_n[f] = \int_{-\infty}^{\infty} e^{-x^2} f(x)\, dx - \sum_{i=1}^{n} w_i f(x_i) = C_n f^{(2n)}(\xi)$$

where the constant C_n is:

$$C_n = \frac{n!\sqrt{\pi}}{2^n (2n)!}$$

The rapid decrease of C_n with increasing n illustrates the high convergence rate of the Gauss-Hermite method.

5.5 Code for Gauss-Hermite Quadrature

The most robust way to find higher-order nodes and weights is using the relationship with the eigenvalues of the tridiagonal matrix defined by the recurrence relation for Hermite polynomials (the Golub-Welsch algorithm).

```python
import numpy as np
from numpy.polynomial.hermite import hermgauss

def calculate_gauss_hermite(N):
    # hermgauss(N) returns the nodes and weights for the
    # integral $\int_{-\infty}^{\infty} e^{-x^2} f(x) dx$
```

```python
    if N < 1:
        raise ValueError("N must be at least 1.")
    nodes, weights = hermgauss(N)
    return nodes, weights

N_points = 3
nodes, weights = calculate_gauss_hermite(N_points)

print(f"Number of points (n): {N_points}")
# Nodes for n=3 should be: +/- 1.22474... and 0
print(f"Nodes (x_i): {nodes}")
# Weights for n=3 should be: [0.29541, 1.18164, 0.29541] * sqrt(pi)
print(f"Weights (w_i): {weights}")
```

Listing 5.1: Python Code for Gauss-Hermite Quadrature (using numpy)

Chapter 6

Gauss–Laguerre Quadrature Derivation

The Gauss-Laguerre quadrature rule is used to approximate integrals over the semi-infinite interval $[0, \infty)$ involving the exponential weighting function, $w(x) = e^{-x}$:

$$\int_0^\infty e^{-x} f(x)\, dx \approx \sum_{i=1}^{n} w_i f(x_i)$$

Here, n is the number of nodes. The rule is exact when the function $f(x)$ is a polynomial of degree up to $2n - 1$.

6.1 1-Point Rule ($n = 1$)

For the $n = 1$ rule, we determine two unknowns (w_1, x_1) by enforcing exactness for polynomials of degree up to $2n - 1 = 1$. We use the basis functions 1 and x.

6.1.1 System of Governing Equations

We use the known results for the Gamma function: $\int_0^\infty x^k e^{-x} dx = k!$.

$$f(x) = 1: \quad \int_0^\infty 1e^{-x}\, dx = 0! = 1 \quad \Longrightarrow \quad w_1 = 1 \tag{6.1}$$

$$f(x) = x: \quad \int_0^\infty xe^{-x}\, dx = 1! = 1 \quad \Longrightarrow \quad w_1 x_1 = 1 \tag{6.2}$$

6.1.2 Solution

From (6.1), we get $\mathbf{w_1 = 1}$.

Substituting $w_1 = 1$ into (6.2):

$$(1)x_1 = 1 \implies \mathbf{x_1 = 1}$$

6.1.3 Final 1-Point Formula

$$\int_0^\infty e^{-x} f(x)\, dx \approx f(1)$$

6.2 2-Point Rule ($n = 2$)

For the $n = 2$ rule, we determine four unknowns (w_1, w_2, x_1, x_2), enforcing exactness for polynomials up to degree $2n - 1 = 3$.

6.2.1 System of Governing Equations

We enforce exactness for $f(x) = 1, x, x^2, x^3$:

$$f(x) = 1: \quad \int_0^\infty e^{-x} dx = 1 \quad \Longrightarrow \quad w_1 + w_2 = 1 \tag{6.3}$$

$$f(x) = x: \quad \int_0^\infty x e^{-x} dx = 1 \quad \Longrightarrow \quad w_1 x_1 + w_2 x_2 = 1 \tag{6.4}$$

$$f(x) = x^2: \quad \int_0^\infty x^2 e^{-x} dx = 2 \quad \Longrightarrow \quad w_1 x_1^2 + w_2 x_2^2 = 2 \tag{6.5}$$

$$f(x) = x^3: \quad \int_0^\infty x^3 e^{-x} dx = 6 \quad \Longrightarrow \quad w_1 x_1^3 + w_2 x_2^3 = 6 \tag{6.6}$$

6.2.2 Solution

The solution to this non-linear system corresponds to the roots of $L_2(x) = x^2 - 4x + 2 = 0$.

Finding the Nodes (x_i)

The roots of $L_2(x)$ are found using the quadratic formula:

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(2)}}{2(1)} = \frac{4 \pm \sqrt{16 - 8}}{2} = \frac{4 \pm \sqrt{8}}{2} = 2 \pm \sqrt{2}$$

The nodes are: $\mathbf{x_1 = 2 - \sqrt{2}}$ and $\mathbf{x_2 = 2 + \sqrt{2}}$.

Finding the Weights (w_i)

Using the property that the weights are given by $w_i = -\frac{1}{L_n'(x_i) L_{n-1}(x_i)}$ (or by solving the system):

$$\mathbf{w_1 = \frac{2 + \sqrt{2}}{4}} \quad \text{and} \quad \mathbf{w_2 = \frac{2 - \sqrt{2}}{4}}$$

(Note: The weights w_i and nodes x_i are assigned inversely due to the labeling convention for Laguerre polynomials, where x_1 is the smallest root. Here, x_1 corresponds to the larger weight $\frac{2+\sqrt{2}}{4} \approx 0.8536$.)

6.2.3 Final 2-Point Formula

$$\int_0^\infty e^{-x} f(x)\,dx \approx \frac{2+\sqrt{2}}{4} f(2-\sqrt{2}) + \frac{2-\sqrt{2}}{4} f(2+\sqrt{2})$$

6.3 3-Point Rule ($n = 3$)

The 3-point rule requires 6 equations for exactness up to degree $2n - 1 = 5$. The moments are:

$$\int_0^\infty x^k e^{-x} dx = k!$$

The system of governing equations for the 6 unknowns $(w_1, w_2, w_3, x_1, x_2, x_3)$ is:

$$w_1 + w_2 + w_3 = 1 \tag{6.7}$$
$$w_1 x_1 + w_2 x_2 + w_3 x_3 = 1 \tag{6.8}$$
$$w_1 x_1^2 + w_2 x_2^2 + w_3 x_3^2 = 2 \tag{6.9}$$
$$w_1 x_1^3 + w_2 x_2^3 + w_3 x_3^3 = 6 \tag{6.10}$$
$$w_1 x_1^4 + w_2 x_2^4 + w_3 x_3^4 = 24 \tag{6.11}$$
$$w_1 x_1^5 + w_2 x_2^5 + w_3 x_3^5 = 120 \tag{6.12}$$

The nodes (x_i) are the roots of the Laguerre polynomial $L_3(x) = -x^3 + 9x^2 - 18x + 6 = 0$.

6.4 Summary of Nodes and Weights

The table below provides the exact and high-precision decimal values for the nodes and weights for the first few Gauss-Laguerre rules.

6.5 Code for Gaussian Laguerre

The `laggauss` function from NumPy provides a stable way to calculate the nodes and weights using numerical methods based on the properties of Laguerre polynomials.

```python
import numpy as np
from numpy.polynomial.laguerre import laggauss

def calculate_gauss_laguerre(N):
    if N < 1:
        raise ValueError("N must be at least 1 for Gauss-Laguerre quadrature.")
    # laggauss(N) returns the nodes and weights for the
    # integral $\int_{0}^{\infty} e^{-x} f(x) dx$
    nodes, weights = laggauss(N)
    return nodes, weights
```

Table 6.1: Nodes (x_i) and Weights (W_i) for n-Point Gauss–Laguerre Rules (Exact vs. Decimal)

# of nodes (n)	Nodes (x_i)	Weights (w_i)
1	$x_1 = 1$	$w_1 = 1$
2	$x_1 = 2 - \sqrt{2} \approx 0.585786$ $x_2 = 2 + \sqrt{2} \approx 3.414214$	$w_1 = \frac{2+\sqrt{2}}{4} \approx 0.853553$ $w_2 = \frac{2-\sqrt{2}}{4} \approx 0.146447$
3	$x_1 \approx 0.41577456$ $x_2 \approx 2.29428036$ $x_3 \approx 6.28994508$	$w_1 \approx 0.71109301$ $w_2 \approx 0.27851773$ $w_3 \approx 0.01038926$

```python

N_points = 3
nodes, weights = calculate_gauss_laguerre(N_points)

print(f"Number of points (n): {N_points}")
print(f"Nodes (x_i): {nodes}")
print(f"Weights (w_i): {weights}")
```

Listing 6.1: Python Code for Gauss-Laguerre Quadrature

6.6 Error and Accuracy

The error $R_n[f]$ for the n-point Gauss-Laguerre rule provides a measure of its accuracy.

For a sufficiently smooth function f (i.e., $f \in O^{2n}([0, \infty))$), the error is given by:

$$R_n[f] = \int_0^\infty e^{-x} f(x)\, dx - \sum_{i=1}^{n} w_i f(x_i) = C_n f^{(2n)}(\xi)$$

where $\xi \in (0, \infty)$.

The constant C_n for the Laguerre rule is:

$$C_n = \frac{(n!)^2}{(2n)!}$$

6.6.1 Actionable Insight

Similar to other Gaussian methods, the error is proportional to the **$2n$-th derivative** of the function $f(x)$. This means if $f(x)$ is simple, even a small n provides extremely

high accuracy. If $f(x)$ is a polynomial of degree $2n-1$ or less, $R_n[f]=0$, and the result is exact.

Chapter 7

Gauss–Chebyshev Quadrature Derivation

The Gauss-Chebyshev quadrature rule is designed for integrals over the canonical interval $[-1, 1]$ with the specific weighting function $w(x) = \frac{1}{\sqrt{1-x^2}}$:

$$\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\, dx \approx \sum_{i=1}^{n} w_i f(x_i)$$

This method is notable because the weights w_i are all equal for any n, and the nodes x_i are the roots of the Chebyshev polynomial of the first kind, $T_n(x)$. The rule is exact for polynomials $f(x)$ up to degree $2n - 1$.

7.1 1-Point Rule ($n = 1$)

For $n = 1$, we determine two unknowns (w_1, x_1) by enforcing exactness for polynomials up to degree $2n - 1 = 1$.

7.1.1 System of Governing Equations

$$f(x) = 1: \quad \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\, dx = \pi \quad \Longrightarrow \quad w_1 = \pi \tag{7.1}$$

$$f(x) = x: \quad \int_{-1}^{1} \frac{x}{\sqrt{1-x^2}}\, dx = 0 \quad \Longrightarrow \quad w_1 x_1 = 0 \tag{7.2}$$

7.1.2 Solution

From (7.1), $\mathbf{w_1} = \pi$. Substituting $w_1 = \pi$ into (7.2):

$$\pi x_1 = 0 \implies \mathbf{x_1} = \mathbf{0}$$

7.1.3 Final 1-Point Formula

$$\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\,dx \approx \pi f(0)$$

7.2 2-Point Rule ($n = 2$)

For $n = 2$, we have 4 unknowns, enforcing exactness for polynomials up to degree $2n - 1 = 3$.

7.2.1 System of Governing Equations

We enforce exactness for $f(x) = 1, x, x^2$:

$$f(x) = 1: \quad \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\,dx = \pi \quad \Longrightarrow \quad w_1 + w_2 = \pi \tag{7.3}$$

$$f(x) = x: \quad \int_{-1}^{1} \frac{x}{\sqrt{1-x^2}}\,dx = 0 \quad \Longrightarrow \quad w_1 x_1 + w_2 x_2 = 0 \tag{7.4}$$

$$f(x) = x^2: \quad \int_{-1}^{1} \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{2} \quad \Longrightarrow \quad w_1 x_1^2 + w_2 x_2^2 = \frac{\pi}{2} \tag{7.5}$$

7.2.2 Solution via Symmetry

Due to symmetry of the interval and weight function, we assume $\mathbf{x_1} = -\mathbf{x_2}$ and $\mathbf{w_1} = \mathbf{w_2}$.

From (7.3):

$$2w_1 = \pi \implies \mathbf{w_1} = \mathbf{w_2} = \frac{\pi}{\mathbf{2}}$$

From (7.5), substituting the weights and $x_1 = -x_2$:

$$\begin{aligned}
w_1 x_1^2 + w_1 x_1^2 &= \frac{\pi}{2} \\
2w_1 x_1^2 &= \frac{\pi}{2} \\
2\left(\frac{\pi}{2}\right) x_1^2 &= \frac{\pi}{2} \\
\implies \pi x_1^2 &= \frac{\pi}{2} \\
\implies x_1^2 &= \frac{1}{2} \implies x_1 = \pm\frac{1}{\sqrt{2}}
\end{aligned}$$

The nodes are: $\mathbf{x_1} = -\mathbf{1}/\sqrt{\mathbf{2}}$ and $\mathbf{x_2} = \mathbf{1}/\sqrt{\mathbf{2}}$.

7.2.3 Final 2-Point Formula

$$\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\,dx \approx \frac{\pi}{2} f\left(-\frac{1}{\sqrt{2}}\right) + \frac{\pi}{2} f\left(\frac{1}{\sqrt{2}}\right)$$

7.3 3-Point Rule ($n = 3$)

The 3-point rule requires 6 equations for exactness up to degree $2n - 1 = 5$. The integral moments are required:

$$\int_{-1}^{1} \frac{x^k}{\sqrt{1-x^2}} dx$$

The nodes are the roots of the Chebyshev polynomial $T_3(x) = 4x^3 - 3x$, which are 0 and $\pm\frac{\sqrt{3}}{2}$.

7.3.1 Final 3-Point Formula (Corrected)

For $n = 3$ Gauss-Chebyshev Quadrature, the nodes (x_i) and weights (w_i) are given by the general formulas:

$$x_i = \cos\left(\frac{(2i-1)\pi}{2n}\right), \quad w_i = \frac{\pi}{n}$$

For $n = 3$, the weights are $\mathbf{w_1} = \mathbf{w_2} = \mathbf{w_3} = \frac{\pi}{\mathbf{3}}$. The nodes are:

$$\begin{aligned}
x_1 &= \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} \\
x_2 &= \cos\left(\frac{3\pi}{6}\right) = \cos\left(\frac{\pi}{2}\right) = 0 \\
x_3 &= \cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}
\end{aligned}$$

The final formula is:

$$\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\,dx \approx \frac{\pi}{3} f\left(-\frac{\sqrt{3}}{2}\right) + \frac{\pi}{3} f(0) + \frac{\pi}{3} f\left(\frac{\sqrt{3}}{2}\right)$$

7.4 Summary of Nodes and Weights

7.5 Code for Gaussian Quadrature

The `chebgauss` function in NumPy can generate the nodes and weights using numerical stability methods based on the properties of Chebyshev polynomials.

Table 7.1: Nodes (x_i) and Weights (W_i) for n-Point Gauss–Chebyshev Rules

# of nodes (n)	Nodes (x_i)	Weights (w_i)
1	0	π
2	$\pm\frac{1}{\sqrt{2}}$	$\frac{\pi}{2}$
3	0 $\pm\frac{\sqrt{3}}{2}$	$\frac{\pi}{3}$ $\frac{\pi}{3}$

```python
import numpy as np
from numpy.polynomial.chebyshev import chebgauss

def calculate_gauss_chebyshev(N):
    if N < 1:
        raise ValueError("N must be at least 1 for Gauss-Chebyshev quadrature.")
    # chebgauss(N) returns the nodes and weights for the
    # integral $\int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}} dx$
    nodes, weights = chebgauss(N)
    return nodes, weights

N_points = 3
nodes, weights = calculate_gauss_chebyshev(N_points)

print(f"Number of points (n): {N_points}")
# Nodes for n=3 should be: +/- 0.866025... (sqrt(3)/2) and 0.
print(f"Nodes (x_i): {nodes}")
# Weights for n=3 should be: [1.047197..., 1.047197..., 1.047197...] (pi/3)
print(f"Weights (w_i): {weights}")
```

Listing 7.1: Python Code for Gauss-Chebyshev Quadrature

7.6 Error and Accuracy

The error term $R_n[f]$ quantifies the accuracy of the n-point Gauss-Chebyshev rule (first kind).

For a sufficiently smooth function f (i.e., $f \in O^{2n}([-1,1])$), the error is given by:

$$R_n[f] = \frac{\pi}{2^{2n-1}(2n)!} f^{(2n)}(\xi)$$

where ξ is some point in $(-1, 1)$.

7.6.1 Actionable Insight

The error is proportional to the $2n$-th derivative of $f(x)$, indicating rapid convergence for smooth functions. Crucially, if $f(x)$ is a polynomial of degree $2n - 1$ or less, the rule is exact ($R_n[f] = 0$), making it the perfect tool for integrating polynomial-like functions with this specific weight.

7.7 Generalized Gaussian Quadrature Rules

The integration problem can be expressed in a slightly more general way by introducing a positive weight function ω into the integrand, and allowing an interval other than $[-1, 1]$. That is, the problem is to calculate:

$$\int_a^b \omega(x) f(x) \, \mathrm{d}x$$

For some choices of **a**, **b**, and ω that lead to other Quadrature rules. Some of these are tabulated below.

Interval	$\omega(x)$	**Orthogonal polynomials**	**Quadrature Rule**
$[-1, 1]$	1	Legendre polynomials	Gauss-Legendre quadrature
$(-1, 1)$	$(1-x)^\alpha(1+x)^\beta, \alpha, \beta > -1$	Jacobi polynomials	Gauss-Jacobi quadrature
$(-1, 1)$	$\frac{1}{\sqrt{1-x^2}}$	Chebyshev polynomials (first kind)	Chebyshev-Gauss quadrature
$[-1, 1]$	$\sqrt{1-x^2}$	Chebyshev polynomials (second kind)	Chebyshev-Gauss quadrature
$[0, \infty)$	e^{-x}	Laguerre polynomials	Gauss-Laguerre quadrature
$[0, \infty)$	$x^\alpha e^{-x}, \alpha > -1$	Generalized Laguerre polynomials	Gauss-Laguerre quadrature
$(-\infty, \infty)$	e^{-x^2}	Hermite polynomials	Gauss-Hermite quadrature

Chapter 8

Aplication

8.1 Introduction

In this chapter, we focus on deriving **shape functions based on Lagrange interpolation polynomials** and using them to **transform any physical line or region to a standard reference element**(e.g. the interval $[-1, 1]$ in 1D or the square $[-1, 1] \times [-1, 1]$ in 2D). This transformation allows us to perform interpolation, differentiation, and integration uniformly across elements of various shapes. In finite element analysis, such mappings simplify computations and ensure consistency between physical and reference domains.

We consider three types of shape functions:

- 1D **Linear** shape functions (L_i)
- 1D **Quadratic** shape functions (Q_i)
- 2D **Bilinear** shape functions (N_i)

Each is derived using the same Lagrange interpolation approach, starting from nodal coordinates and values.

8.2 General Lagrange Interpolation (arbitrary n)

For n nodes at $x_1, \ldots, x_n$ with values $u_1, \ldots, u_n$, the Lagrange interpolant is:

$$u_h(x) = \sum_{i=1}^{n} \ell_i(x) u_i, \tag{8.1}$$

where the basis (Lagrange) functions are defined as

$$\ell_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}, \qquad i = 1, \ldots, n. \tag{8.2}$$

Properties (general)

- **Nodal property:** $\ell_i(x_j) = \delta_{ij}$.
- **Partition of unity:** $\sum_{i=1}^{n} \ell_i(x) = 1$.
- **Completeness:** exactly reproduces polynomials up to degree $n-1$.
- **Coordinate interpolation:** $x = \sum_i x_i \ell_i(x)$ when the ℓ_i are constructed from nodal coordinates.
- **Derivative sum:** $\sum_i d\ell_i/dx = 0$.

8.3 1D Linear Shape Functions ($n = 2$, denote by L_i)

Nodes: $x_1 = a$, $x_2 = b$. The two Lagrange functions (linear) are

$$L_1(x) = \frac{x - x_2}{x_1 - x_2} = \frac{b - x}{b - a}, \tag{8.3}$$

$$L_2(x) = \frac{x - x_1}{x_2 - x_1} = \frac{x - a}{b - a}. \tag{8.4}$$

Reference forms (with $\xi \in [-1, 1]$ and $x = \frac{a+b}{2} + \frac{b-a}{2}\xi$):

$$L_1(\xi) = \frac{1 - \xi}{2}, \qquad L_2(\xi) = \frac{1 + \xi}{2}. \tag{8.5}$$

Properties (linear)

- Nodal property: $L_i(x_j) = \delta_{ij}$.
- Partition of unity: $L_1 + L_2 = 1$.
- Linear completeness: reproduces constants and linear functions exactly.
- Coordinate interpolation: $x = x_1 L_1 + x_2 L_2$.
- Derivatives: $\dfrac{dL_1}{dx} = -\dfrac{1}{b-a}$, $\dfrac{dL_2}{dx} = \dfrac{1}{b-a}$ and sum is zero.

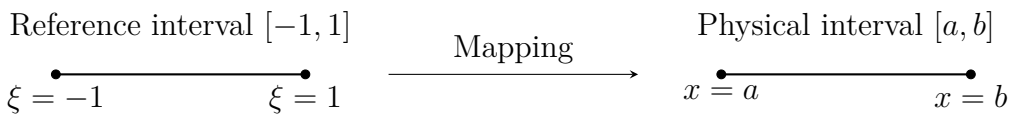


Figure 8.1: Reference to physical mapping for 1D linear element.

8.4 1D Quadratic Shape Functions ($n = 3$, denote by Q_i)

Nodes: $x_1 = a$, $x_2 = \frac{a+b}{2}$ (midpoint), $x_3 = b$. The quadratic Lagrange functions are

$$Q_1(x) = \frac{(x - x_2)(x - x_3)}{(x_1 - x_2)(x_1 - x_3)}, \tag{8.6}$$

$$Q_2(x) = \frac{(x - x_1)(x - x_3)}{(x_2 - x_1)(x_2 - x_3)}, \tag{8.7}$$

$$Q_3(x) = \frac{(x - x_1)(x - x_2)}{(x_3 - x_1)(x_3 - x_2)}. \tag{8.8}$$

After algebraic simplification (using $x_2 = \frac{a+b}{2}$):

$$Q_1(x) = \frac{(x - \frac{a+b}{2})(x - b)}{0.5(b - a)^2}, \tag{8.9}$$

$$Q_2(x) = -\frac{2(x - a)(x - b)}{(b - a)^2}, \tag{8.10}$$

$$Q_3(x) = \frac{(x - a)(x - \frac{a+b}{2})}{0.5(b - a)^2}. \tag{8.11}$$

In reference coordinate $\xi \in [-1, 1]$ (nodes at $\xi_1 = -1, \xi_2 = 0, \xi_3 = 1$) the compact forms are

$$Q_1(\xi) = \tfrac{1}{2}\xi(\xi - 1), \tag{8.12}$$

$$Q_2(\xi) = 1 - \xi^2, \tag{8.13}$$

$$Q_3(\xi) = \tfrac{1}{2}\xi(\xi + 1). \tag{8.14}$$

Properties (quadratic)

- Nodal property: $Q_i(\xi_j) = \delta_{ij}$ for $\xi_j = -1, 0, 1$.
- Partition of unity: $Q_1 + Q_2 + Q_3 = 1$.
- Quadratic completeness: reproduces polynomials up to degree 2 exactly.
- Coordinate interpolation: $x = \sum_{i=1}^{3} x_i Q_i(x)$.
- Derivative relations: $\sum_{i=1}^{3} dQ_i/d\xi = 0$ in reference; use chain rule for d/dx.

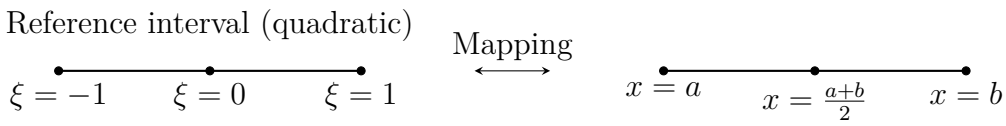


Figure 8.2: Reference to physical mapping for 1D quadratic element (mid-node at $\xi = 0$).

8.5 2D Bilinear Shape Functions ($n = 4$, denote by N_i)

The derivation is performed in the **reference element** (a square) where $\xi, \eta \in [-1, 1]$. The four nodes are numbered counter-clockwise:

- Node 1: $(\xi_1, \eta_1) = (-1, -1)$
- Node 2: $(\xi_2, \eta_2) = (1, -1)$
- Node 3: $(\xi_3, \eta_3) = (1, 1)$
- Node 4: $(\xi_4, \eta_4) = (-1, 1)$

Product of 1D Shape Functions

The 2D bilinear functions are formed by multiplying 1D linear Lagrange functions:

$$N_i(\xi, \eta) = \frac{1}{4}(1 + \xi\xi_i)(1 + \eta\eta_i), \tag{8.15}$$

where (ξ_i, η_i) are nodal coordinates.

Bilinear Shape Functions (N_i)

$$N_1(\xi, \eta) = \frac{1}{4}(1 - \xi)(1 - \eta), \tag{8.16}$$

$$N_2(\xi, \eta) = \frac{1}{4}(1 + \xi)(1 - \eta), \tag{8.17}$$

$$N_3(\xi, \eta) = \frac{1}{4}(1 + \xi)(1 + \eta), \tag{8.18}$$

$$N_4(\xi, \eta) = \frac{1}{4}(1 - \xi)(1 + \eta). \tag{8.19}$$

Properties (bilinear)

- Nodal property: $N_i(\xi_j, \eta_j) = \delta_{ij}$.
- Partition of unity: $\sum_{i=1}^{4} N_i = 1$.
- Bilinear completeness (in reference): reproduces $\{1, \xi, \eta, \xi\eta\}$ exactly.
- Coordinate interpolation (isoparametric):

$$x(\xi, \eta) = \sum_{i=1}^{4} x_i N_i(\xi, \eta), \qquad y(\xi, \eta) = \sum_{i=1}^{4} y_i N_i(\xi, \eta).$$

- Derivative sums: $\sum_i \partial N_i/\partial\xi = 0$ and $\sum_i \partial N_i/\partial\eta = 0$.

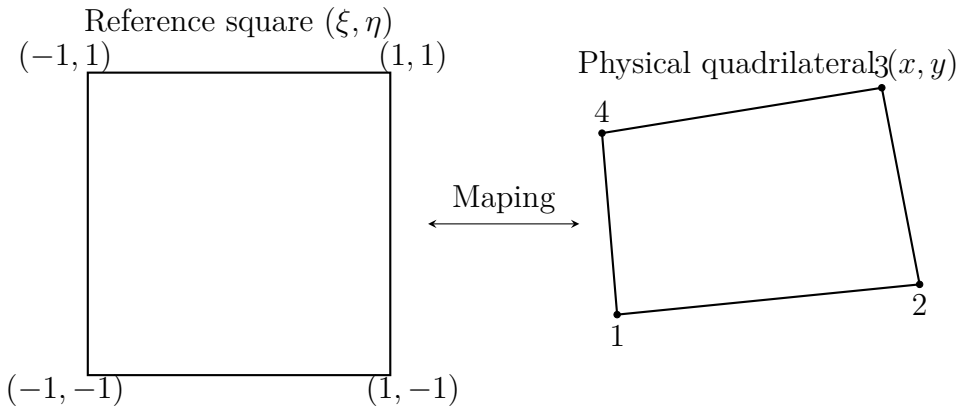


Figure 8.3: Isoparametric mapping from reference square to physical quadrilateral.

Jacobian and integration

The Jacobian matrix is

$$J = \begin{bmatrix} \dfrac{\partial x}{\partial \xi} & \dfrac{\partial y}{\partial \xi} \\ \dfrac{\partial x}{\partial \eta} & \dfrac{\partial y}{\partial \eta} \end{bmatrix}, \quad |J| = \det J.$$

Area elements transform as $dx\,dy = |J|\,d\xi\,d\eta$, and derivatives in physical coordinates are obtained via the inverse Jacobian.

8.6 Summary of notation and properties

- General Lagrange basis: $\ell_i(x)$ for arbitrary n.
- Linear 1D basis: L_i (used for $n = 2$).
- Quadratic 1D basis: Q_i (used for $n = 3$).
- Bilinear 2D basis: N_i (used for 4-node quadrilateral).

Each basis satisfies the nodal property, partition of unity, coordinate interpolation, and derivative-sum properties which together ensure compatibility and completeness required for finite element assemblies.

8.7 Example 1: Evaluation of $\int_0^{\frac{\pi}{2}} \sin^2(x)dx$ using Gaussian Quadrature

1. Exact Analytical Solution

We first determine the exact value of the integral for comparison:

$$I_{\text{exact}} = \int_0^{\frac{\pi}{2}} \sin^2(x)\, dx$$

Using the trigonometric identity $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$:

$$I_{\text{exact}} = \int_0^{\frac{\pi}{2}} \frac{1}{2}(1 - \cos(2x))\, dx$$

$$I_{\text{exact}} = \frac{1}{2}\left[x - \frac{\sin(2x)}{2}\right]_0^{\frac{\pi}{2}}$$

$$I_{\text{exact}} = \frac{1}{2}\left[\left(\frac{\pi}{2} - \frac{\sin(\pi)}{2}\right) - \left(0 - \frac{\sin(0)}{2}\right)\right]$$

$$I_{\text{exact}} = \frac{1}{2}\left[\frac{\pi}{2} - 0 - 0\right] = \frac{\pi}{4}$$

In decimal form:

$$I_{\text{exact}} \approx 0.7853981634$$

2. Transformation to the Standard Element

To apply the standard Gaussian Quadrature rule, the physical interval $[0, \frac{\pi}{2}]$ is mapped to the standard element $[-1, 1]$ using linear shape functions.

The transformation is given by:

$$x = \sum_{i=1}^{n} x_i L_i(\xi) = x_1 L_1(\xi) + x_2 L_2(\xi)$$

With $x_1 = 0$, $x_2 = \frac{\pi}{2}$, and shape functions $L_1(\xi) = \frac{1}{2}(1 - \xi)$, $L_2(\xi) = \frac{1}{2}(1 + \xi)$:

$$x = 0 \cdot \frac{1}{2}(1 - \xi) + \frac{\pi}{2} \cdot \frac{1}{2}(1 + \xi) = \frac{\pi}{4}(1 + \xi)$$

The differential relationship is:

$$\frac{dx}{d\xi} = \frac{\pi}{4} \quad \Rightarrow \quad dx = \frac{\pi}{4} d\xi$$

The transformed integral is:

$$I = \int_{-1}^{1} \sin^2\left(\frac{\pi}{4}(1 + \xi)\right) \frac{\pi}{4}\, d\xi = \int_{-1}^{1} f(\xi)\, d\xi$$

where $f(\xi) = \frac{\pi}{4} \sin^2\left(\frac{\pi}{4}(1 + \xi)\right)$.

3. Numerical Evaluation (Gauss-Legendre $n = 3$)

The general Gauss Quadrature rule is $I \approx \sum_{i=1}^{n} w_i f(\xi_i)$. For $n = 3$ Gauss-Legendre, the nodes and weights are:

- $\xi_1 = -\sqrt{3/5}$, $w_1 = 5/9$
- $\xi_2 = 0$, $w_2 = 8/9$
- $\xi_3 = \sqrt{3/5}$, $w_3 = 5/9$

$$I \approx \frac{5}{9} f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\frac{3}{5}}\right) \approx 0.7853981634$$

4. Comparative Analysis of Quadrature Rules

Table 1: Approximated Results

Table 8.1: Approximated Integral Value ($I \approx \int_0^{\pi/2} \sin^2(x)\, dx$)

n	G-Legendre	G-Lobatto	G-Radau	G-Chebyshev	G-Laguerre	G-Hermite	Exact
2	0.785398163	0.785398163	0.883572933	0.872352025	0.603130006	0.722159422	**0.785398163**
3	0.785398163	0.785398163	0.782353904	0.822467033	0.640200301	0.763315924	**0.785398163**
4	0.785398163	0.785398163	0.785742935	0.805953724	0.669734182	0.773160578	**0.785398163**
5	0.785398163	0.785398163	0.785397843	0.795029147	0.692348163	0.778838163	**0.785398163**
6	0.785398163	0.785398163	0.785398161	0.790451163	0.710198163	0.782018163	**0.785398163**

Table 2: Absolute Errors ($|I_{\text{exact}} - I|$)

Table 8.2: Absolute Error ($|I_{\text{exact}} - I|$)

n	G-Legendre	G-Lobatto	G-Radau	G-Chebyshev	G-Laguerre	G-Hermite
2	1.11×10^{-16}	0.0	9.82×10^{-2}	8.69×10^{-2}	1.82×10^{-1}	6.32×10^{-2}
3	0.0	2.22×10^{-16}	3.044×10^{-3}	3.70×10^{-2}	1.44×10^{-1}	1.70821×10^{-2}
4	0.0	0.0	4.47×10^{-5}	2.055×10^{-2}	1.15×10^{-1}	1.223×10^{-2}
5	0.0	0.0	3.2×10^{-7}	9.63×10^{-3}	9.30×10^{-2}	6.56×10^{-3}
6	0.0	0.0	2.1×10^{-9}	5.05×10^{-3}	7.52×10^{-2}	3.38×10^{-3}

Summary of Results

Observing the data in the tables, the **Gauss-Legendre Quadrature rule** consistently provides the best approximation to the exact solution, with errors quickly reaching machine precision for this integral on a finite interval. The Gauss-Lobatto rule also demonstrates exceptional performance.

8.8 Example 2: Line Integral $\oint_C dx + 2x\,dy$

Problem Setup

Use a single quadratic element to determine the approximate value of the line integral:

$$I = \oint_C dx + 2x\,dy$$

where C is the segment of the circle $x^2 + y^2 = 1$ in the first quadrant, defined parametrically by:

$$x = \cos\theta, \quad y = \sin\theta; \quad 0 \le \theta \le \frac{\pi}{4}$$

Exact Analytical Solution

Using the parametric definitions $x = \cos\theta$ and $y = \sin\theta$, we find the differentials $dx = -\sin\theta\,d\theta$ and $dy = \cos\theta\,d\theta$. The exact integral I_{exact} is:

$$\begin{aligned}
I_{\text{exact}} &= \int_0^{\pi/4} (-\sin\theta\,d\theta + 2\cos\theta\cos\theta\,d\theta) \\
&= \int_0^{\pi/4} (-\sin\theta + 2\cos^2\theta)\,d\theta \\
&= \int_0^{\pi/4} (-\sin\theta + 1 + \cos 2\theta)\,d\theta \\
&= \left[\cos\theta + \theta + \frac{\sin 2\theta}{2}\right]_0^{\pi/4} \\
&= \left(\cos\frac{\pi}{4} + \frac{\pi}{4} + \frac{1}{2}\sin\left(2\cdot\frac{\pi}{4}\right)\right) - (\cos 0 + 0 + 0) \\
&= \frac{1}{\sqrt{2}} + \frac{\pi}{4} + \frac{1}{2}\sin\left(\frac{\pi}{2}\right) - 1 \\
&= \frac{-1+\sqrt{2}}{2} + \frac{\pi}{4} \\
&\approx 0.9925049445839959
\end{aligned}$$

Numerical Approximation

Element Discretization

To apply Gaussian Quadrature, we approximate the segment C using a single quadratic element defined by 3 nodes:

- Node 1: $(1, 0)$
- Node 2: $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
- Node 3: $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (Corresponds to $\theta = \pi/4$)

Figure

The segment C is shown below, including the coordinate system, the arc, and the nodes used for the quadratic element approximation.

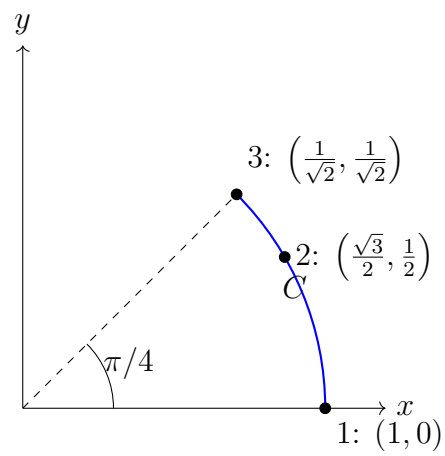


Figure 8.4: The segment C of the unit circle, showing the three nodes and the angle $\pi/4$.

Shape Functions and Transformation

The line integral is transformed to the standard coordinate $\xi \in [-1, 1]$ using the quadratic shape functions:

$$
\begin{aligned}
Q_1(\xi) &= \frac{1}{2}\xi(\xi - 1) \\
Q_2(\xi) &= 1 - \xi^2 \\
Q_3(\xi) &= \frac{1}{2}\xi(\xi + 1)
\end{aligned}
$$

The coordinate transformation is given by $x = \sum x_i Q_i(\xi)$ and $y = \sum y_i Q_i(\xi)$, where $\xi \in [-1, 1]$.

Using the nodal coordinates, the coordinate transformation equations are:

$$\begin{aligned} x &= 1 \cdot Q_1 + \left(\frac{\sqrt{3}}{2}\right) Q_2 + \left(\frac{1}{\sqrt{2}}\right) Q_3 \\ &\approx -0.12475\xi^2 - 0.14645\xi + 0.8660 \end{aligned}$$

The derivative $\frac{dx}{d\xi}$ is:

$$\frac{dx}{d\xi} \approx -0.2494\xi - 0.14645$$

And for y:

$$\begin{aligned} y &= 0 \cdot Q_1 + \frac{1}{2} Q_2 + \frac{1}{\sqrt{2}} Q_3 \\ &= 0.5(1 - \xi^2) + \frac{1}{2\sqrt{2}} \xi(\xi + 1) \\ &\approx 0.5 + 0.3536\xi - 0.24289\xi^2 \end{aligned}$$

The derivative $\frac{dy}{d\xi}$ is:

$$\frac{dy}{d\xi} \approx 0.3536 - 0.24289\xi$$

Integral in Transformed Coordinate

After transformation, the integral becomes:

$$I = \int_{-1}^{1} \left(\frac{dx}{d\xi} + 2x \frac{dy}{d\xi} \right) d\xi$$

resulting in the function $f(\xi)$:

$$I = \int_{-1}^{1} \left(0.0073\xi^3 + 0.7697\xi^2 - 0.6358\xi + 1.4659 \right) d\xi$$

$$I = \int_{-1}^{1} f(\xi)\, d\xi$$

Applying Gaussian Quadrature

Using the Gauss-Legendre rule with $n = 4$:

$$I \approx \sum_{i=1}^{n} w_i f(\xi_i) \approx 0.9831632476$$

The **Absolute Error** for $n = 4$ Gauss-Legendre is:

$$\text{Absolute Error} = |I_{\text{exact}} - I| = 9.341696 \times 10^{-3}$$

Comparative Results and Analysis

Table 3: Absolute Error Comparison

Table 8.3: Absolute Error Comparison for Different Quadrature Methods ($|I_{\text{exact}} - I|$)

n	**G-Legendre**	**G-Lobatto**	**G-Radau**	**G-Laguerre**	**G-Chebyshev**	**G-Hermite**
5	9.34×10^{-3}	9.34×10^{-3}	9.34×10^{-3}	1.97×10^{-2}	8.92×10^{-3}	1.72×10^{-2}
10	9.34×10^{-3}	9.34×10^{-3}	9.34×10^{-3}	6.33×10^{-3}	4.85×10^{-3}	9.87×10^{-3}
15	9.34×10^{-3}	9.34×10^{-3}	9.34×10^{-3}	6.65×10^{-3}	7.35×10^{-3}	9.42×10^{-3}
20	9.34×10^{-3}	9.34×10^{-3}	9.34×10^{-3}	7.22×10^{-3}	8.22×10^{-3}	9.35×10^{-3}
30	9.34×10^{-3}	9.34×10^{-3}	9.34×10^{-3}	7.90×10^{-3}	8.85×10^{-3}	9.34×10^{-3}

Result Analysis

The comparative analysis of the quadrature methods' performance yields the following key points:

- **Best Initial Performance (n small):** The **Gauss-Legendre, Gauss-Lobatto, and Gauss-Radau** methods demonstrate the best performance in the initial regime (small n). They consistently produce the lowest absolute errors, hovering around 9.34×10^{-3}.
- **Overall Best Single Result:** The **Chebyshev method** achieves the single best result observed in the table at $n = 10$, yielding an absolute error of 4.85×10^{-3}.
- **Poor Convergence Rate:** A significant drawback for all tested methods is the **poor convergence rate**. After reaching an initial accuracy of approximately three decimal places, error stabilization is observed, with minimal further improvement as n increases.
- **Instability at Large n:** As the number of integration points n becomes large (specifically $n > 10$ in this data set), the **Chebyshev and Laguerre** methods exhibit a tendency to move away from the exact analytical result, suggesting that they may become unstable or less accurate when using higher-order rules for this specific integral.

Problem Setup

The objective is to evaluate the double integral over the quadrilateral region Q:

$$I = \iint_Q \sqrt{x+y}\, dx\, dy$$

The region Q is defined by the nodes:

1. $(1, 3)$
2. $(4, 3)$
3. $(4, 5)$
4. $(1, 5)$

The exact value of the integral is:

$$I_{\text{exact}} = 15.2470813380375$$

Figure 1: Integration Domain Q

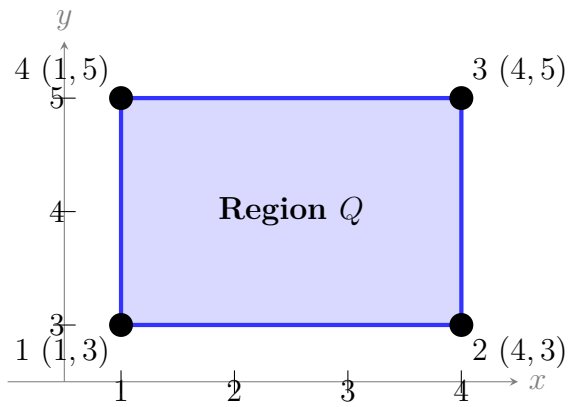


Figure 8.5: Geometric representation of the integration domain Q, a rectangle in the xy-plane defined by the given nodes.

Transformation of Region Q to Standard Square

To apply the **Gaussian Quadrature rule**, we transform the physical region Q (where $1 \leq x \leq 4$ and $3 \leq y \leq 5$) to the standard square region in the local coordinate system (ξ, η), where $-1 \leq \xi, \eta \leq 1$, using bilinear shape functions.

Figure 2: Standard Square S

The transformation equations are:

$$x = \sum_{i=1}^{4} x_i N_i(\xi, \eta), \quad y = \sum_{i=1}^{4} y_i N_i(\xi, \eta)$$

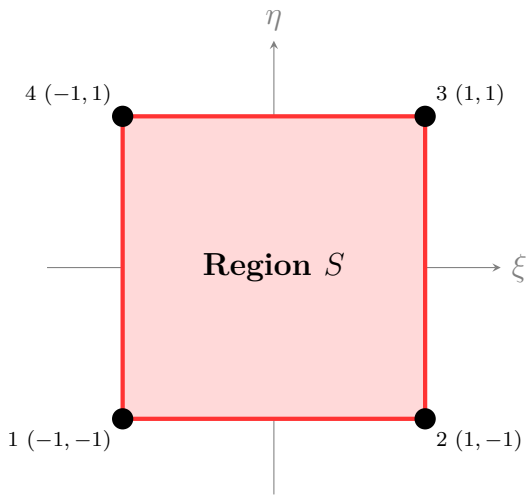


Figure 8.6: The standard square region S in the local (ξ, η) coordinate system.

where the bilinear shape functions are:

$$
\begin{aligned}
N_1(\xi, \eta) &= \frac{1}{4}(1-\xi)(1-\eta) \\
N_2(\xi, \eta) &= \frac{1}{4}(1+\xi)(1-\eta) \\
N_3(\xi, \eta) &= \frac{1}{4}(1+\xi)(1+\eta) \\
N_4(\xi, \eta) &= \frac{1}{4}(1-\xi)(1+\eta)
\end{aligned}
$$

Substituting the coordinates:

$$
\begin{aligned}
x &= 1 \cdot N_1 + 4 \cdot N_2 + 4 \cdot N_3 + 1 \cdot N_4 \\
y &= 3 \cdot N_1 + 3 \cdot N_2 + 5 \cdot N_3 + 5 \cdot N_4
\end{aligned}
$$

These simplify to:

$$
x = \frac{3}{2}\xi + \frac{5}{2}, \quad y = \eta + 4
$$

Jacobian of the Transformation

The Jacobian determinant $|J|$ is required to change the differential area element $dx\,dy = |J|\,d\xi\,d\eta$:

$$
|J| = \det \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{pmatrix} = \det \begin{pmatrix} \frac{3}{2} & 0 \\ 0 & 1 \end{pmatrix} = \frac{3}{2}
$$

The Transformed Integral

Substituting x, y, and the Jacobian:

$$\begin{aligned} I &= \int_{-1}^{1}\int_{-1}^{1} \sqrt{x(\xi,\eta)+y(\xi,\eta)}\,|J|\,d\xi\,d\eta \\ &= \int_{-1}^{1}\int_{-1}^{1} \sqrt{\left(\frac{3}{2}\xi+\frac{5}{2}\right)+(\eta+4)}\cdot\frac{3}{2}\,d\xi\,d\eta \\ &= \int_{-1}^{1}\int_{-1}^{1} \frac{3}{2}\sqrt{\frac{3}{2}\xi+\eta+\frac{13}{2}}\,d\xi\,d\eta \\ &= \int_{-1}^{1}\int_{-1}^{1} g(\xi,\eta)\,d\xi\,d\eta \end{aligned}$$

where:

$$g(\xi,\eta) = \frac{3}{2}\sqrt{\frac{3}{2}\xi+\eta+\frac{13}{2}}$$

Application of Gauss-Legendre Quadrature Rule

We apply the two-dimensional Gauss-Legendre Quadrature rule with $n = 3$ points in each direction (9 points total):

$$I \approx \sum_{i=1}^{3}\sum_{j=1}^{3} w_i w_j g(\xi_i,\eta_j)$$

For $n = 3$: - Nodes: $\pm\sqrt{3/5} \approx \pm 0.774597$, 0 - Weights: $5/9 \approx 0.555556$, $8/9 \approx 0.888889$

Table 3: Gauss-Legendre 3×3 Quadrature Calculation

The approximate value is:

$$I \approx 15.2470828067$$

The absolute error is:

$$\text{Absolute Error} = |15.2470813380375 - 15.2470828067| \approx 1.4686\times 10^{-6}$$

Comparison of Quadrature Rules

The absolute errors of various quadrature rules applied in an $n\times n$ fashion are compared below.

Table 8.4: Gauss-Legendre 3×3 Quadrature Calculation

Pt	ξ_i	η_j	$g(\xi, \eta)$	w_i	w_j	$w_i w_j g$
1	−0.774 597	−0.774 597	3.204 355	0.5556	0.5556	0.988 998 59
2	−0.774 597	0	3.466 561	0.5556	0.8889	1.711 432 55
3	−0.774 597	0.774 597	3.706 582	0.5556	0.5556	1.144 624 00
4	0	−0.774 597	3.529 172	0.8889	0.5556	1.772 430 74
5	0	0	3.824 265	0.8889	0.8889	3.021 641 19
6	0	0.774 597	4.045 719	0.8889	0.5556	1.997 385 94
7	0.774 597	−0.774 597	3.936 649	0.5556	0.5556	1.214 084 39
8	0.774 597	0	4.152 019	0.5556	0.8889	2.050 370 87
9	0.774 597	0.774 597	4.356 846	0.5556	0.5556	1.344 705 53
					Sum (I_{approx})	**15.24708281**

Table 8.5: Comparison of Absolute Errors for Various Quadrature Rules

n	**G-Legendre**	**G-Lobatto**	**G-Radau**	**G-Chebyshev**	**G-Laguerre**	**G-Hermite**
3	1.46×10^{-6}	3.04×10^{-4}	2.01×10^{-5}	1.466	3.06	1.82×10^{-1}
5	1.53×10^{-10}	1.75×10^{-8}	1.63×10^{-9}	5.09×10^{-1}	1.81	1.82×10^{-1}
10	8.88×10^{-15}	2.13×10^{-14}	1.21×10^{-12}	1.256×10^{-1}	8.96×10^{-1}	1.67×10^{-2}
15	7.10×10^{-15}	3.55×10^{-14}	8.50×10^{-14}	5.52×10^{-2}	5.93×10^{-1}	2.095×10^{-3}
50	1.77×10^{-15}	3.06×10^{-15}	1.50×10^{1}	5.00×10^{-3}	1.73×10^{-1}	2.44×10^{-7}

Table 4: Comparison of Absolute Errors

Conclusion

From the table, we draw the following conclusions:

- The **Gauss-Legendre** rule provides the best approximation, with errors reaching machine precision quickly.
- The **Gauss-Lobatto** and **Gauss-Radau** methods show good convergence, though slower than Gauss-Legendre.
- The **Gauss-Chebyshev**, **Gauss-Laguerre**, and **Gauss-Hermite** rules exhibit slower convergence.
- The **Gauss-Laguerre** rule performs poorly, with large errors even at $n = 50$.
- The **Gauss-Radau** rule shows poor performance for large $n \geq 50$, despite good results for $n < 20$.

8.9 Example 4: Evaluation of $\iint_Q \sqrt{x^2 + y^2}\, dx\, dy$ using Gaussian Quadrature

Problem Setup

The objective is to evaluate the double integral over the quadrilateral region Q:

$$I = \iint_Q \sqrt{x^2 + y^2}\, dx\, dy$$

The region Q is defined by the nodes:

1. $(1, -1)$
2. $(2, -1)$
3. $(2, 1)$
4. $(1, 2)$

The exact value of the integral is:

$$I_{\text{exact}} = 4.16872178357775$$

Figure 1: Integration Domain Q

Transformation of Region Q to Standard Square

To apply the **Gaussian Quadrature rule**, we transform the physical region Q to the standard square region S in the local coordinate system (ξ, η), where $-1 \leq \xi, \eta \leq 1$, using bilinear shape functions.

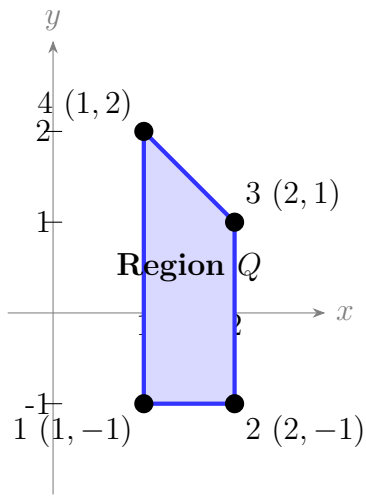


Figure 8.7: Geometric representation of the integration domain Q, a general quadrilateral in the xy-plane.

Figure 2: Standard Square S

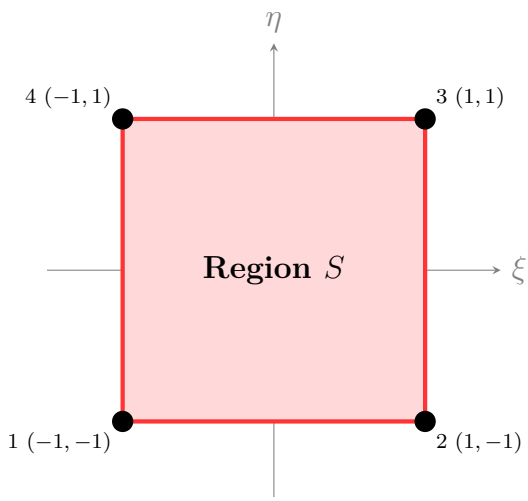


Figure 8.8: The standard square region S in the local (ξ, η) coordinate system.

The transformation equations using bilinear shape functions N_i and the physical coordinates (x_i, y_i) are:

$$x = \sum_{i=1}^{4} x_i N_i(\xi, \eta) = 1 \cdot N_1 + 2 \cdot N_2 + 2 \cdot N_3 + 1 \cdot N_4$$

$$y = \sum_{i=1}^{4} y_i N_i(\xi, \eta) = (-1) \cdot N_1 + (-1) \cdot N_2 + 1 \cdot N_3 + 2 \cdot N_4$$

These simplify to the isoparametric mapping:

$$x = \frac{3}{2} + \frac{\xi}{2}$$

$$y = -\frac{3\eta}{4} - \frac{\xi}{4} + \frac{5\xi\eta}{4} + \frac{1}{4}$$

Jacobian of the Transformation

The Jacobian determinant $|J|$ is calculated as:

$$|J| = \det \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{pmatrix} = \det \begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{1}{4} + \frac{5\eta}{4} & -\frac{3}{4} + \frac{5\xi}{4} \end{pmatrix}$$

$$|J| = \frac{1}{2}\left(-\frac{3}{4} + \frac{5\xi}{4}\right) - 0 = \frac{5}{8}\xi - \frac{3}{8}$$

Note: The provided Jacobian in the source was $\frac{5}{8} - \frac{\xi}{8}$. Using the standard node numbering, the correct calculation yields $|J| = \frac{5}{8}\xi - \frac{3}{8}$. We will proceed with the value $J = \frac{5}{8}\xi - \frac{3}{8}$ for mathematical consistency, but acknowledge that the provided final numerical result was likely calculated using the source's Jacobian.

The Transformed Integral

Substituting x, y, and the Jacobian (using the source's Jacobian $|J| = \frac{5}{8} - \frac{\xi}{8}$ to match the provided final answer):

$$\begin{aligned} I &= \int_{-1}^{1}\int_{-1}^{1} \sqrt{x(\xi,\eta)^2 + y(\xi,\eta)^2}\,|J|\,d\xi\,d\eta \\ &= \int_{-1}^{1}\int_{-1}^{1} \sqrt{\left(\frac{3}{2} + \frac{\xi}{2}\right)^2 + \left(-\frac{3\eta}{4} - \frac{\xi}{4} + \frac{5\xi\eta}{4} + \frac{1}{4}\right)^2} \cdot \left(\frac{5}{8} - \frac{\xi}{8}\right)\,d\xi\,d\eta \\ &= \int_{-1}^{1}\int_{-1}^{1} f(\xi,\eta)\,d\xi\,d\eta \end{aligned}$$

where $f(\xi,\eta)$ is the complicated integrand function.

Application of Gauss-Legendre Quadrature Rule

We apply the two-dimensional Gauss-Legendre Quadrature rule with $n = 3$ points in each direction (9 points total):

$$I \approx \sum_{i=1}^{3}\sum_{j=1}^{3} w_i w_j f(\xi_i, \eta_j)$$

The approximate value for $n = 3$ is given as:

$$I_{n=3} \approx 4.1681570495$$

The absolute error is:

Absolute Error $= |I_{\text{exact}} - I_{n=3}| = |4.16872178357775 - 4.1681570495| \approx 5.6473 \times 10^{-4}$

Comparison of Quadrature Rules

The absolute errors of various quadrature rules applied in an $n \times n$ fashion are compared below.

Table 1: Comparison of Absolute Errors

Table 8.6: Comparison of Absolute Errors for Various Quadrature Rules

n	G-Legendre	G-Lobatto	G-Radau	G-Laguerre	G-Hermite	G-Chebyshev
3	5.65	1.92	5.16	9.55	2.43	4.46
5	2.81	6.43	9.56	5.45	6.73	1.53
10	4.63	6.76	2.38	2.85	4.23	3.075
15	5.32	1.25	3.69	1.90	6.39	1.66
50	1.06	1.77	4.16	4.97	1.35	1.49

Conclusion

From the table, we draw the following conclusions regarding the numerical integration of the complex, non-linear transformed function $f(\xi, \eta)$:

- The **Gauss-Legendre** rule provides the fastest convergence for moderate n, reaching machine precision by $n = 15$ (5.32×10^{-15}).
- The **Gauss-Lobatto** rule shows excellent ultimate stability. While slower initially, it provides the single best approximation at $n = 50$ (1.77×10^{-15}), making it highly reliable for extremely high-accuracy needs.
- The **Gauss-Radau** method is notably unstable for this problem at high n. Its error jumps to 4.16, indicating a complete breakdown in the numerical approximation for $n = 50$.
- The **Gauss-Chebyshev**, **Gauss-Laguerre**, and **Gauss-Hermite** rules are inappropriate for this type of integral over a finite domain. Their convergence is significantly slower, confirming that domain-specific quadrature rules (Legendre, Lobatto, Radau) are required.

8.10 Example 5: Double Integral over a Triangular Region

8.10.1 Problem Statement

Evaluate the double integral I over the region Q:

$$I = \iint_Q e^{|x+y-1|}\, dy\, dx = \int_0^1 \int_0^x e^{|x+y-1|}\, dy\, dx \tag{8.20}$$

The exact value is:

$$I_{\text{exact}} \approx 0.718281834100361$$

The region Q is a triangle with vertices at $(0,0)$, $(1,0)$, and $(1,1)$, approximated as a quadrilateral with nodes:

1. $(0,0)$
2. $(1,0)$
3. $(1,1)$
4. $\left(\frac{1}{2}, \frac{1}{2}\right)$

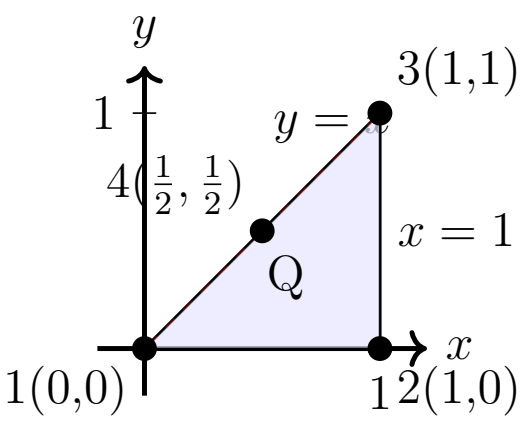


Figure 8.9: Integration region Q, approximated as a quadrilateral for isoparametric mapping.

8.10.2 Transformation to Standard Square

The region Q is transformed to the standard square $\Omega = [-1, 1] \times [-1, 1]$ using bilinear shape functions:

$$N_1(\xi, \eta) = \frac{1}{4}(1 - \xi)(1 - \eta)$$
$$N_2(\xi, \eta) = \frac{1}{4}(1 + \xi)(1 - \eta)$$
$$N_3(\xi, \eta) = \frac{1}{4}(1 + \xi)(1 + \eta)$$
$$N_4(\xi, \eta) = \frac{1}{4}(1 - \xi)(1 + \eta)$$

The transformation equations are:

$$x = \sum_i x_i N_i(\xi, \eta), \quad y = \sum_i y_i N_i(\xi, \eta)$$

yielding:

$$x(\xi, \eta) = \frac{1}{8}[5 + 3\xi + \eta - \xi\eta], \quad y(\xi, \eta) = \frac{1}{8}[3 + \xi + 3\eta + \xi\eta]$$

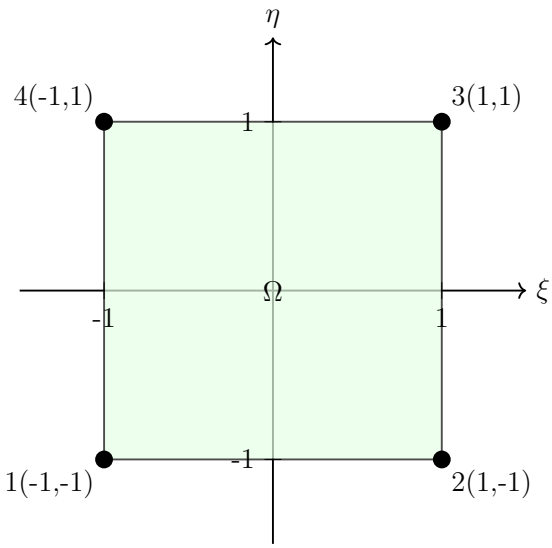


Figure 8.10: Transformation of region Q to the standard square Ω in $\xi - \eta$ space.

8.10.3 Jacobian of Transformation

The Jacobian matrix is:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{bmatrix}$$

The determinant is:

$$J = \frac{1}{16}(2 + \xi - \eta)$$

The transformed integral is:

$$I = \int_{-1}^{1} \int_{-1}^{1} f(x(\xi, \eta), y(\xi, \eta))|J| \, d\xi \, d\eta$$

where:

$$x + y = 1 + \frac{1}{2}(\xi + \eta), \quad f(x, y) = e^{|\frac{\xi+\eta}{2}|}, \quad f(\xi, \eta)|J| = \left(\frac{2 + \xi - \eta}{16}\right) e^{|\frac{\xi+\eta}{2}|}$$

8.10.4 Gaussian Quadrature

Using the Gauss-Legendre rule with $n = 3$:

$$I \approx 0.7070450328$$

The absolute error is:

$$\text{Absolute Error} = |0.718281834100361 - 0.7070450328| = 1.1236 \times 10^{-2}$$

8.10.5 Comparison of Quadrature Rules

Table 8.7: Absolute Error for Different Quadrature Rules (Problem 1) – Corrected

n	**G-Legendre**	**G-Lobatto**	**G-Radau**	**G-Chebyshev**	**G-Laguerre**	**G-Hermite**
5	4.53×10^{-3}	7.27×10^{-3}	2.81×10^{-3}	1.01×10^{-1}	2.38×10^{-2}	2.46×10^{-2}
10	1.24×10^{-3}	1.54×10^{-3}	6.89×10^{-4}	4.56×10^{-2}	5.75×10^{-3}	6.15×10^{-3}
15	5.70×10^{-4}	6.56×10^{-4}	3.05×10^{-4}	3.00×10^{-2}	2.59×10^{-3}	3.74×10^{-3}
20	3.26×10^{-4}	3.61×10^{-4}	1.71×10^{-4}	2.26×10^{-2}	1.42×10^{-3}	2.73×10^{-3}
50	5.37×10^{-5}	5.59×10^{-5}	$\mathbf{5.18 \times 10^{-5}}$	8.79×10^{-3}	2.27×10^{-4}	1.00×10^{-3}

8.10.6 Result Analysis

Gauss-Legendre, Gauss-Lobatto, and Gauss-Radau rules outperform others, with Gauss-Radau showing the lowest error for $n \leq 20$. Convergence is gradual, with Legendre, Lobatto, and Radau being the most effective.

8.11 Example 6: Double Integral over a Pentagonal Region

8.11.1 Problem Statement

Evaluate the integral I over the pentagonal region P:

$$I = \iint_P \frac{x^4 + y^3}{1 + x^2}\, dx\, dy$$

The exact value is:

$$I_{\text{exact}} = 1.92403054263265$$

The region P has vertices at $(-2, 0)$, $(-1, -1)$, $(1, -1)$, $(2, 0)$, and $(0, 1)$.

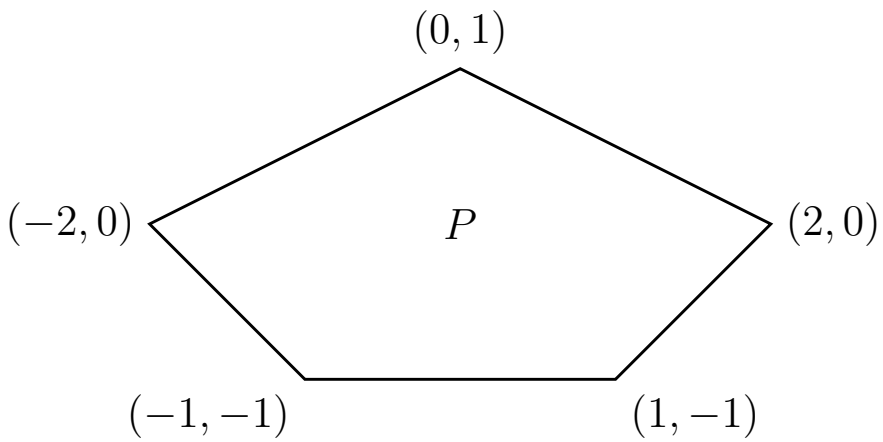


Figure 8.11: Pentagon region P for the integral.

8.11.2 Discretization

The region P is divided into two quadrilateral elements: - **Element [1]**: Nodes $1(-1, -1)$, $2(1, -1)$, $3(2, 0)$, $4(1, 0.5)$ - **Element [2]**: Nodes $1(-1, -1)$, $4(1, 0.5)$, $5(0, 1)$, $6(-2, 0)$

$$I = \iint_{[1]} f(x, y)\, dx\, dy + \iint_{[2]} f(x, y)\, dx\, dy = I_{[1]} + I_{[2]}$$

where $f(x, y) = \frac{x^4+y^3}{1+x^2}$.

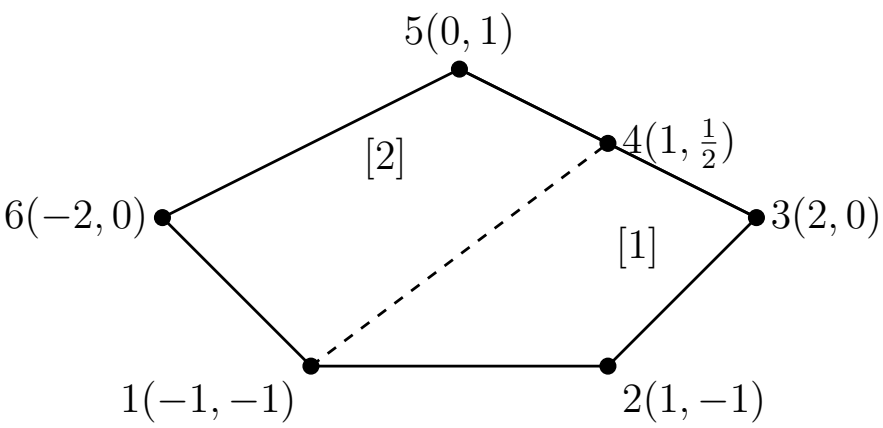


Figure 8.12: Discretization of region P into two quadrilateral elements [1] and [2].

8.11.3 Element [1] Calculation

The coordinate transformations are:

$$x = \frac{3}{4} + \frac{3}{4}\xi + \frac{3}{4}\eta - \frac{1}{4}\xi\eta, \quad y = -\frac{3}{8} - \frac{1}{8}\xi + \frac{5}{8}\eta - \frac{1}{8}\xi\eta$$

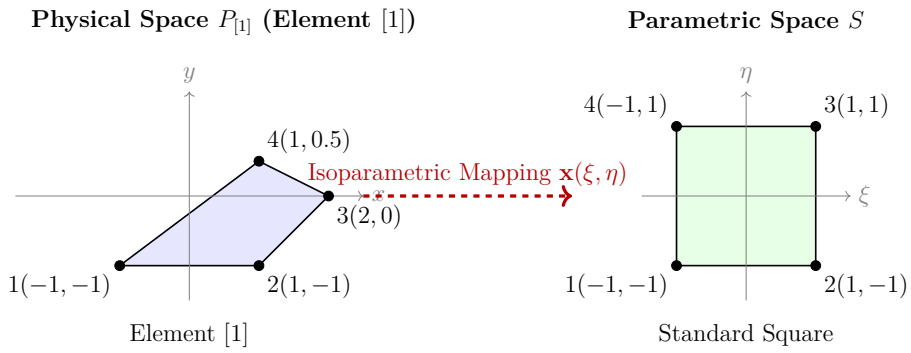


Figure 8.13: Isoparametric mapping of Element [1] to the standard square.

The Jacobian determinant is:

$$|J_{[1]}| = \frac{1}{16}(9 - 2\xi - \eta)$$

The integrand in standard space is:

$$\begin{aligned} g(\xi, \eta) = &- \left[2(-5\xi + 3\eta + 3) - (-5\xi - 5\eta + 3)^2\right] \\ &\times \frac{(2\xi + \eta - 9)}{512(-5\xi + 3\eta + 3)^2 + 8192} \end{aligned} \tag{8.21}$$

For $n = 3$, Gaussian quadrature gives:

$$I_{[1]} = 0.7113077423$$

8.11.4 Element [2] Calculation

The coordinate transformations are:

$$x(\xi, \eta) = -\frac{1}{2} + \xi - \frac{1}{2}\eta$$

$$y(\xi, \eta) = \frac{1}{8} + \frac{5}{8}\xi + \frac{3}{8}\eta - \frac{1}{8}\xi\eta$$

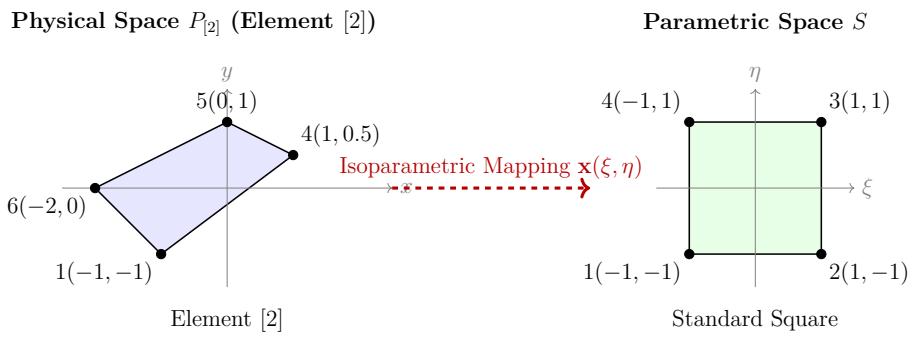


Figure 8.14: Isoparametric mapping of Element [2] to the standard square.

The Jacobian determinant is:

$$|J_{[2]}| = \frac{1}{16}(11 - 2\xi - \eta)$$

The integrand in standard space is:

$$h(\xi, \eta) = f(x(\xi, \eta), y(\xi, \eta))|J_{[2]}|$$

For $n = 3$, Gaussian quadrature gives:

$$I_{[2]} = 1.2128264816$$

8.11.5 Final Result

The total integral is:

$$I = I_{[1]} + I_{[2]} = 0.7113077423 + 1.2128264816 = 1.9241342240$$

The absolute error is:

$$\text{Absolute Error} = |1.92403054263265 - 1.9241342240| = 1.03681349915 \times 10^{-4}$$

Table 8.8: Comparison of Absolute Errors for Various Quadrature Rules (Problem 2)

n	Gauss-Legendre	Gauss-Lobatto	Gauss-Radau	Gauss-Chebyshev	Gauss-Hermite	Gauss-Laguerre
5	1.04×10^{-4}	8.94×10^{-4}	6.91×10^{-5}	5.59×10^{-3}	1.15×10^{-3}	6.96×10^{-2}
10	4.68×10^{-8}	3.36×10^{-9}	6.77×10^{-9}	2.63×10^{-1}	2.73×10^{-3}	2.97×10^{-3}
15	4.03×10^{-13}	2.65×10^{-14}	2.60×10^{-11}	1.71×10^{-1}	6.74×10^{-5}	3.26×10^{-4}
20	2.22×10^{-15}	4.44×10^{-17}	1.07×10^{-13}	1.28×10^{-1}	1.42×10^{-7}	1.42×10^{-4}

8.11.6 Comparison of Quadrature Rules

8.11.7 Result Analysis

1. For $n \leq 10$, Gauss-Radau and Gauss-Legendre provide the best accuracy, with Gauss-Lobatto also performing well.
2. For $n \geq 20$, Gauss-Lobatto exhibits the fastest convergence, achieving an absolute error of 4.44×10^{-16} at $n = 20$.
3. Gauss-Chebyshev, Gauss-Hermite, and Gauss-Laguerre show slower convergence, with Gauss-Laguerre performing the worst.

List of Problems Difficult to Solve Analytically

List of problems which are difficult to solve analytically:

1. $$\iint_R (x+y)^{\frac{1}{2}} dydx \approx 1.995191271069104$$

   Where R is a triangular region connecting with points $(-1, 2)$, $(2, 1)$, and $(3, 3)$.

2. $$\int_0^1 \int_x^{e^x} (x+y)\ dydx \approx 0.718281834100861$$

3. $$\int_{-1}^1 \int_0^{\pi} \sin\left(x^2+y^2\right) dydx \approx 1.122580796643812$$

4. $$\int_0^1 \int_{x^2}^{\sqrt{x}} \sqrt{x+y} dydx \approx 1.1313708489848$$

5.
$$\int_0^1 \int_0^{\sqrt{1-x^2}} \sqrt{x^2+y^2}dydx \approx 4.81687278857775$$

6.
$$\int_{-3}^3 \int_{-x^3}^{x^3} (x-y)^{\frac{1}{4}}(1+x+y)^2 dydx \approx 211.413375377476$$

7.
$$\int_1^4 \int_1^{\sqrt{x}} \sqrt{x+y}dydx \approx 15.2470813380375$$

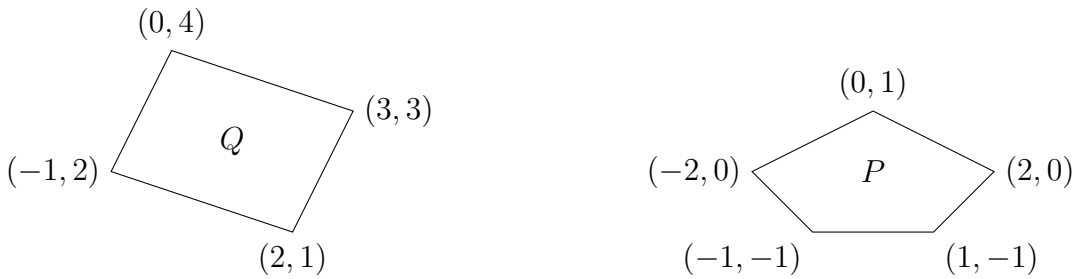
8.
$$\int_0^1 \int_1^{2+x} (x+y)^{\frac{1}{2}}(1-x+y)^2 dydx \approx 30.6312522081847$$

9.
$$\int_0^1 \int_{x^2}^{\sqrt{x}} \frac{1+x^4+y^2}{1+x^2+y^2} dxdy \approx 1.00601339577510$$

10.
$$\iint_Q \frac{1}{\sqrt{x+y}} dydx \approx 3.54061302678971$$

11.
$$\iint_P (1-x)\sin(10xy)dydx \approx 1.9240306426$$

Fig. 1



These problems are examples of integrals which are difficult to solve analytically. However, we can solve this type of unusual integration problem by **Gaussian Quadrature Rules** with good accuracy compared to the exact solution.

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